

PROM1-associated retinal degeneration

An allele-aware evidence map of mechanisms, interventions and research priorities

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A review of the published evidence on PROM1-associated retinal degeneration, covering what each study showed in relation to which allele, which cell compartment and which stage of the disease. It also sets out seven comparisons that would help characterise those differences more accurately, and so assist in the design of treatments and trials.

If you have work on PROM1 in the retina, published or not, or think something here is wrong, the author would be grateful to hear from you.

Abstract

PROM1-associated retinal degeneration encompasses dominant macular disease and recessive, often more widespread photoreceptor degeneration. Their distinct inheritance patterns and clinical presentations raise different mechanistic and therapeutic questions. This review and evidence map integrates clinical genetics, natural history, membrane biology, photoreceptor and retinal pigment epithelium (RPE) models, preclinical interventions and dated human-study records. The strongest translational advance is rescue in PROM1-deficient experimental systems, demonstrated in patient-derived organoids and in knockout mice. Post-onset benefit is reported in a full-text mouse study and in a separate conference abstract. The principal unresolved transfer is to dominant mutant disease, where augmentation, mutant suppression and combined treatment require direct comparison. The mechanistic literature extends beyond outer-segment morphogenesis: direct RPE experiments implicate autophagic flux and outer-segment clearance, while light-sensitive knockout models implicate endothelin, insulin-like growth factor 1 (IGF1) and metabolic stress. These pathways require compartment-specific interpretation. Human quantitative autofluorescence findings constrain a universal bisretinoid-excess explanation, and protective local endothelin effects in other retinal genotypes argue against treating the pathway as uniformly harmful. Clinical endpoints are correspondingly stratified: dominant atrophy expansion, recessive ellipsoid-zone contraction and layer-specific thinning on optical coherence tomography (OCT) measure different processes and have different floor effects. Human studies with explicit or broader genetic eligibility may admit PROM1 disease, but for most of them the records show who could enter rather than who did, or what followed. Seven experimental programmes connect these distinctions to testable comparisons, each set out as the decisive contrast, the safeguard it needs, and the decision it would settle. The resulting framework distinguishes candidate therapies by allele, compartment and disease stage.

Keywords: PROM1; prominin-1; CD133; STGD4; MCDR2; CORD12; RP41; inherited retinal degeneration; macular dystrophy; autophagic flux; gene augmentation; retinal pigment epithelium.

1. Introduction

Prominin was described in 1997 by Weigmann, Corbeil, Hellwig and Huttner. A monoclonal antibody they raised against mouse neuroepithelium identified a glycoprotein with an extracellular N-terminus, five transmembrane segments, two large glycosylated extracellular loops and a cytoplasmic C-terminus. In epithelia it sat on the apical surface, selectively associated with microvilli; expressed in Chinese hamster ovary (CHO) cells it was preferentially localised to filopodia, lamellipodia and microspikes rather than the neighbouring flat membrane, and they concluded that it carries the information that targets it to, or keeps it in, membrane protrusions rather than the planar cell surface. In the same year Yin, Miraglia and colleagues used a separate antibody, AC133, to identify a marker of human haematopoietic stem and progenitor cells. [84,85]

The two lines met in 2000, when Maw and colleagues linked the gene to human retinal degeneration. Their report describes it as human prominin-like 1, PROML1, previously known as the AC133 antigen, which is why the protein is also called CD133 and why much of its literature concerns stem cells and cancer rather than the eye. Their report followed from where the protein sits and from its expression in retinoblastoma cell lines and adult retina, not from the haematopoietic work. Photoreceptor disks are produced by outgrowths of the plasma membrane, so a gene encoding a retinal protein targeted to plasma membrane protrusions was a candidate for inherited retinal degeneration; that is the reasoning of the 2000 report. It studied a consanguineous Indian pedigree in which an uncle–niece marriage produced eight children, four of them affected, who reported night blindness and loss of peripheral vision from childhood and had extinguished electroretinogram (ERG) responses by their third decade. The affected members were homozygous for a deletion at nucleotide 1878, which shifts the reading frame from codon 614 and terminates translation after twelve amino acids unrelated to the protein. The predicted product would lack about half of the second extracellular loop, the last membrane-spanning segment and the cytoplasmic C-terminal domain. The deletion was absent from 97 Indian controls, and two-point analysis treating it as a rare polymorphism gave a peak logarithm-of-odds (LOD) score of 3.17 at zero recombination. A mouse prominin construct carrying a similar deletion, expressed in CHO cells, did not reach the cell surface, and mouse prominin itself was concentrated in the plasma membrane evaginations at the base of rod outer segments. The authors' hypothesis was that losing prominin causes retinal degeneration by impairing the generation of those evaginations, or their conversion into disks, or both. Yang and colleagues then showed that mutant prominin 1 from patients with macular degeneration disrupts photoreceptor disk morphogenesis in mice, and subsequent work established the dominant p.Arg373Cys phenotype more broadly. The clinical spectrum now includes macular dystrophy, bull's-eye maculopathy, cone–rod disease and retinitis-pigmentosa-like presentations. These labels describe overlapping patterns rather than one uniform mechanism or trajectory. [1–6]

The therapeutic question is consequently whether the proposed intervention restores the relevant function in the relevant allele class, cell compartment and disease stage. A deficient photoreceptor may benefit from replacement of a missing protein; a cell containing a disruptive mutant protein may require a different balance of replacement and suppression. Preserved photoreceptors, compromised RPE and an extensively atrophic macula also present different biological targets. Recent gene-delivery studies make these distinctions more actionable. [7–10]

PROM1-RELATED RETINAL DISEASE

How membrane disorganisation and impaired cellular maintenance can lead to vision loss

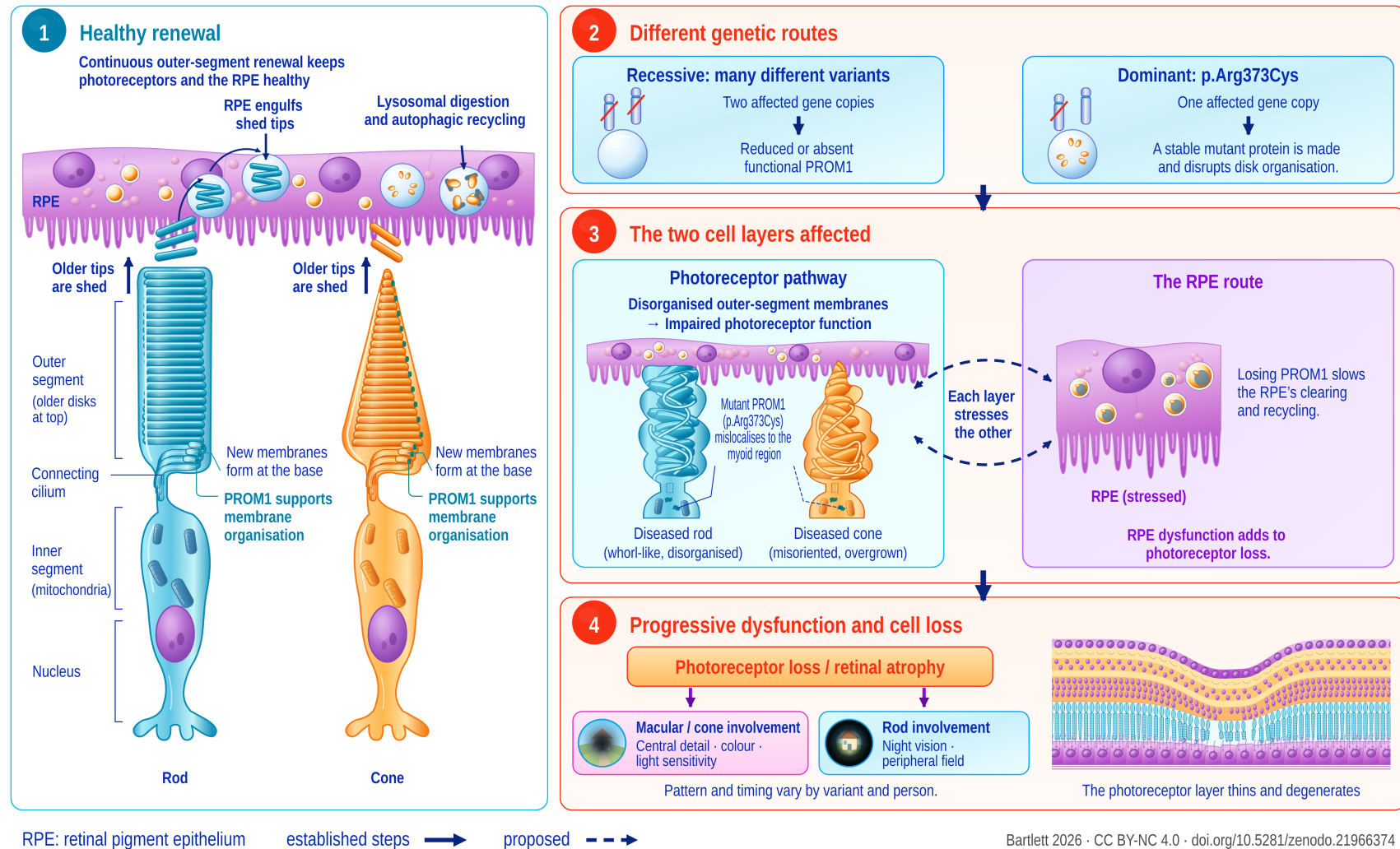


Figure 1. PROM1-related retinal disease. A working model of the condition as currently understood: outer-segment renewal and what disrupts it (1, 2), the two cell compartments affected (3), and the loss of vision that follows (4). Dashed lines mark what is proposed rather than established. The evidence for each element is in Sections 3 to 10.

This review follows a chain from allele and transcript to protein abundance, localisation and membrane behaviour; from cellular renewal and clearance to tissue injury; and from experimental rescue to a clinically measurable outcome. Its central purpose is to identify where an inference is supported directly, where it transfers across models, and which experiment would most efficiently resolve the remaining uncertainty. The clinical and experimental tables are evidence maps, not pooled estimates or treatment recommendations.

Bhattacharya and colleagues synthesised the expansion of *PROM1* biology from photoreceptor outer-segment structure to retinal pigment epithelium homeostasis, linking autophagy, lysosomal clearance and epithelial state to a proposed connection with atrophic age-related macular degeneration. Their review sets out a dual-compartment model and discusses allele-specific suppression, replacement and combined-compartment strategies. The present review develops the inherited-disease evidence at the level of allele, model construction, assay, compartment and treatment stage. It examines the transfer from deficient-system rescue to endogenous dominant disease, compares light and pharmacological interventions, and connects dated human-study records with quantitative natural-history endpoints. Seven proposed experimental programmes specify comparisons, confounders and falsifying outcomes. This complements the biological synthesis with a detailed account of what individual results establish about intervention choice and timing. [11]

The stem-cell line of that history became a large literature of its own. CD133 is used as a marker of tumour-initiating cells across several cancers, and of the papers that name the gene or the protein, more than half concern cancer while about one in forty concerns the retina. None of that work is covered here.

1.1 Severity, course and functional impact

PROM1 disease is phenotypically heterogeneous, with a wide range of outcomes. At the severe end, biallelic disease takes away central vision in childhood or early adult life. In Cehajic-Kapetanovic and colleagues' Oxford and Bonn series, individuals carrying recessive variants commonly had severe loss of visual acuity by their twenties; they describe the condition as a severely blinding disease. In Yates and colleagues' Australian longitudinal series, in six patients followed for a median of 11.8 years, acuity held up until after the age of twelve to fifteen and then fell steeply, with symptoms first noticed at about seven. Structural loss ran ahead of the acuity loss a chart records by three to five years. Measured acuity across Lin and colleagues' Taiwanese series ran from 20/20 down to hand motion at five centimetres. Ragi and colleagues reported compound heterozygous frameshift variants, c.139del and c.1877_1878del, in one child given a clinical diagnosis of Leber congenital amaurosis: nystagmus and photodysphoria from the age of two, poor vision at three, and at twelve a bull's-eye maculopathy with an extinguished electroretinogram. They describe it as the first case of *PROM1*-associated Leber congenital amaurosis. A Brazilian screen of 43 families diagnosed with Leber congenital amaurosis or early-onset severe retinal dystrophy, published two years earlier, reported two probands homozygous for *PROM1* splice variants, c.784+1G>T and c.2130+2del, and does not state which of the two clinical labels applied to either. Only one of the two is counted as solved by *PROM1*: that proband appears in the screen's table of families solved by genes not typically associated with Leber congenital amaurosis. The other carries its homozygous *PROM1* variant alongside two heterozygous *CRB1* variants, and the screen counts that family among those solved by a canonical Leber congenital amaurosis gene, so the *PROM1* allele there is an additional finding rather than the assigned cause. Two probands assigned to

PROM1 across the two reports, only one of them with a recorded clinical label, mark the edge of the reported range rather than a described subgroup. [5,18,58,79,109]

What that costs is severe. Central vision is the vision used for faces, print, screens and fine work. Losing it in the second or third decade arrives during education and the start of working life, and it removes driving, most print reading and a large part of the usual route into employment. Blurred vision was the commonest symptom in the Taiwanese series, in eight of its fourteen patients, while photophobia, distorted vision and impaired colour vision were each reported by one, and the single Leber congenital amaurosis case had slight photodysphoria from the age of two. Contrast sensitivity is not reported in any of these series, so the burden beyond acuity is documented in individual patients rather than counted across them. Night blindness, the presenting complaint of two of the seven patients in Hwang and colleagues' Korean cohort and reported by one of the fourteen in the Taiwanese one, restricts independent movement after dark. Where vision is reduced from early childhood, as in that single case, the same losses arrive during schooling rather than in adult life, though not to the same degree: the child's best-corrected acuity was 20/80 and 20/70 at twelve, which is substantially reduced central acuity rather than its absence. [13,18,79]

The dominant p.Arg373Cys phenotype is generally milder and later, and many carriers retain good acuity into late adulthood, in part because the fovea is often spared: seven of ten patients in Fujinami and colleagues' Japanese series, 70%, had a spared fovea to some degree, and foveal sparing was what maintained acuity in those who kept it. That relative mildness is real, and it varies: the same variant produces different trajectories in different people and sometimes within one family, so an individual prognosis cannot be read off the genotype. Reported penetrance differs between series, and the evidence is allele-specific. The Taiwanese cohort describes asymptomatic heterozygous carriers of c.139del and of c.794del. It suggests that incomplete penetrance could explain the c.139del carriers and extends the suggestion to dominant *PROM1* alleles, while for the two asymptomatic c.794del sons it considers late onset or an unknown modifier and leaves that allele's pathogenicity unsettled. For the recurrent p.Arg373Cys allele the same paper records that an earlier series reported complete penetrance and raises incomplete penetrance only as a possibility, because not every unaffected relative had imaging and genotyping. A three-family dominant series reports high penetrance and consistent segregation. So the documented unaffected carriers carry other alleles, and whether p.Arg373Cys is fully penetrant remains open. [4,5,6,18,62]

Onset can be later still. Among 22 Korean adults whose cone or cone-rod dystrophy began after the age of 30, *PROM1* was one of six genes carrying a disease-causing variant, although RP1, CRX and CDHR1 accounted for 77% of that series. Whether the phenotype reaches beyond the eye has been examined directly. Two kindreds carrying p.Arg373Cys were examined for vascular and endothelial measures on the reasoning that CD133 marks circulating progenitor cells, and aspects of endothelial function assayed outside the body were abnormal, with impaired adhesion capacity and higher levels of cellular damage; renal infections, haematuria and recurrent miscarriages were also noted. The same study examined the brain, on the reasoning that prominin-1 has a putative role in hippocampal neurogenesis. Hippocampal and cerebral volumes were within normal ranges in both kindreds and no distinctive pattern of cognitive deficit emerged, but three carriers had small white-matter lesions, one carrier in each family showed some memory and executive impairment, three members of one family had an empty sella turcica with normal baseline pituitary function, and three carriers had impaired smell; the authors read these findings, with the endothelial ones, as suggesting endothelial dysfunction that could become more evident with age. The authors close by saying

further studies are needed to confirm the positive findings. No confirming study was found in the sources assembled here, so *PROM1* disease is treated as ocular throughout, with the endothelial, renal and neurological observations recorded as raised and unconfirmed. [99,100]

1.2 Relative frequency, distribution and the treatment position

The disease is rare, which shapes the research problem. *PROM1* accounted for 37 of 3,195 molecularly diagnosed families, 1.2%, in the largest United Kingdom inherited-retinal-disease cohort. Rarity means small series, few eligible participants for any trial, and cohorts assembled across countries with different ascertainment. It does not mean the mechanism is obscure: outer-segment renewal, retinal pigment epithelium clearance and photoreceptor survival are shared with commoner disease. *PROM1*-deficient models have also been linked to RPE processes relevant to atrophic age-related macular degeneration, a connection developed in the review by Bhattacharya and colleagues and examined here in Section 4.4, on RPE injury in vivo. [11,14,53]

PROM1 disease has been reported from Europe, Asia, the Middle East, Africa, North America, South America and Australasia, and the reported series are concentrated where inherited-retinal-disease genetic testing is organised rather than where the gene is necessarily commonest. No African series was found in the sources assembled here, which is a statement about this literature rather than about the gene; Habibi and colleagues report one Tunisian family with retinitis pigmentosa and homozygous c.1354dupT, within a fifteen-family cohort covering many genes. That is the allele corrected in the patient-derived stem cells of Section 3.2 [12,110]. Pontikos and colleagues' United Kingdom service cohort contributes 37 families and 53 affected individuals, 1.2% of that cohort on both counts [14]. Series have been published from Japan [6], Korea [13], China [15], India [17], Taiwan [18], Australia [58], Spain [60] and Italy [61], and a 49-patient Indian Stargardt cohort included one patient with a *PROM1* variant [16]; an Israeli family supplied one of the three reported noncoding alleles [73], and a consanguineous Pakistani pedigree supplied one of the founding recessive descriptions [76]. In a São Paulo macular-dystrophy series, Salles and colleagues reported three unrelated patients carrying *PROM1* variants among 25 records; two variants were proposed as disease-associated and the third case remained molecularly unresolved [111]. The Brazilian screen of 43 families is described in Section 1.1 [109]. Section 3.3 sets out why these counts cannot be combined into a prevalence estimate.

Cohorts sequenced across many genes at once answer a different question, because they carry a denominator. Table 9 sets out 46 of them, each testing 100 or more people or families, and gives the *PROM1* count wherever the cohort states one. Where a cohort states both a solved total and a *PROM1* count, the share ranges about tenfold: 4 of 1,091 solved in the Australian registry, 3 of 574 in a German diagnostic series, 20 of 2,036 in one Italian centre, 9 of 400 in São Paulo and 17 of 400 in western Switzerland, the last of which its authors give as 4.3%. In the largest single collection, 377 of 33,834 patients submitted with a reported genetic cause carried *PROM1*. These cohorts differ in whom they enrolled, in what they counted and in whether the denominator is everyone tested or everyone solved, and several draw on the same clinical services, so the figures are set beside each other rather than pooled. [123,126–128,131,134]

One pattern in those series is consistent enough to matter for trial design. The recurrent dominant allele p.Arg373Cys was carried by all seven patients from all five Korean families reported [13], accounted for six of the eight families in the Japanese national series and six of nine, 66.7%, across the literature review accompanying it [6], and appeared again among

the Taiwanese [18] and Chinese [15] variants. A single recurrent allele that concentrates in populations reached by organised ascertainment, from the Japan Eye Genetics Consortium's nationwide series to single-centre Chinese and Korean cohorts, is a workable starting position for a genotype-specific study: the eligible population is identifiable, the allele is the same one in each participant, and the ascertainment pathway already exists.

The therapeutic work has followed. Both *PROM1* gene-replacement programmes published as journal articles are Chinese, one from Zhongshan Ophthalmic Center in Guangzhou and one from a Shanghai collaboration, and the disclosed vector patents are Chinese. East Asia therefore holds several of the best-characterised p.Arg373Cys cohorts and both of those vector programmes, which is an argument about where the assembled infrastructure is and not a claim that the allele belongs there: the same variant is reported from the United Kingdom, Spain and Italy. Trial-ready populations for the biallelic forms are more dispersed, which is one reason the natural-history work in Section 9 and the registry record set out in Section 7 and Table 5 matter as much as the vector work. [6,8,9,13–15,18,28,29,58,63,72,73]

No approved treatment for any form of *PROM1*-associated retinal degeneration was identified in the regulatory, registry and sponsor sources reviewed to 18 September 2026. Those records show particular programmes and their status; a formal census of every regulator was not attempted. [34–40]

2. Scope, source handling and inference rules

2.1 Review design

This is an independent review and evidence map: no sponsor, no institutional affiliation and no commercial interest in any of the programmes it describes. It is written as a narrative synthesis with structured evidence tables, and it appraises each claim against the study that supports it. It is not a protocol-driven systematic review or meta-analysis: no prospective protocol was registered, PRISMA systematic-review procedures were not applied, and no effect estimate is pooled across studies.

The literature was found by searching *PROM1*, prominin-1 and CD133 against retinal, photoreceptor, macular, Stargardt, cone-rod, retinitis pigmentosa and retinal pigment epithelium terms, then by chasing citations from what that returned and by reading trial-registry, sponsor and patent records directly. Citation chasing followed disputed quantitative claims, direct *PROM1* experiments, clinical cohorts, natural-history endpoints and trial eligibility. English-language indexing was supplemented by searching in Chinese, Japanese, Korean, Russian, Spanish, Portuguese, German and Turkish, and by the Latin American databases SciELO and LILACS. Some closed databases remain unaccessed, so for example the Chinese and Japanese literature here is mostly what is openly indexed. Searching was iterative: an unresolved claim sent it back to the literature. A study in which *PROM1* is one subgroup among many is not usually titled for it, and Table 9 collects those studies. Everything cited was published on or before 18 September 2026, and the regulatory, registry and sponsor position is stated as it stood on that date.

Verification was claim-specific. Published full texts and supplements were preferred for methods, denominators and numerical results; a primary abstract supported only the information it actually reported. Review articles were used as maps to primary work rather than counted as independent experiments. A preprint and its later publication were not counted twice. Access is therefore stated rather than assumed. This review cites 162 sources. The full published text

was read for 133 of them. Fifteen are not journal articles or preprints but trial registry entries, sponsor announcements and study protocols. Eight are known only from an abstract or a supplement. Six could not be obtained, and wherever a claim rests on one of those, it says so. Appendix A places every source in one of these groups. Every paragraph that carries a citation, and every row of the nine tables, has since been checked sentence by sentence against the sources it cites. Claims resting on the six sources that could not be obtained were not checked this way. The author retains a source-access and review-disposition record, distinguishing first-hand inspection, inherited extraction, abstract-level evidence and unresolved leads. No new patient-level meta-analysis, sequence analysis, population-genetic test or transcriptomic reanalysis was performed. Neither is safety assessed. Every intervention described here is weighed for whether it has been shown to work and in what system, not for whether it can be given safely, and that includes the delivery route: the *PROM1* gene-replacement experiments reviewed in vivo used subretinal injection, whose risks in a macula that still has a spared fovea are a separate question this review does not take up.

2.2 Units, versions and biological interpretation

Evidence is organised by allele, construct, compartment, stage, assay and biological unit. An eye is not automatically an independent animal; an organoid or clone is not automatically an independent donor; thousands of single cells do not replace biological replication. Clinical reports are not summed without accounting for overlapping families, repeated publications and ascertainment. When a paper's abstract and body differ, both descriptions are retained in the source audit and the more granular accounting is used where justified.

Reported HGVS strings are preserved as each source reports them. A protein-level shorthand does not establish a transcript accession, genome build, phase or molecular consequence. The illustrative variant table therefore separates a source-reported allele from what was measured experimentally. It is not a newly normalised clinical variant database. Likewise, a knockout, viral knockdown, overexpression transgene and endogenous knock-in are distinct models even when each is labelled "*PROM1* disease".

For human programmes, registry status is tied to a dated record; sponsor information is separately attributed. Formal eligibility, practical screening eligibility, actual genotype composition, subgroup outcomes and regulatory milestones are distinct fields. A passed estimated start date does not itself change "not yet recruiting" to "recruiting", and acceptance of a regulatory application is not approval.

3. Clinical genetics

3.1 Dominant and recessive disease

Table 1 sets out the illustrative allele classes used here and what each establishes. The recurrent p.Arg373Cys allele is an important dominant disease model, but its phenotype is not restricted to an isolated macular lesion. Clinical reports describe differing macular and peripheral involvement, including peripheral bone-spicule degeneration in a Chinese cohort. Retinal phenotype and stage are therefore part of what stratifies dominant disease, alongside the variant. Alongside that clinical variation there is a molecular observation. Wild-type prominin-1 sits at the nascent disk membranes at the base of the outer segment; in Yang and colleagues' mutant transgenic lines the protein was found most strongly in the myoid region of the inner segment, which is largely endoplasmic reticulum, and scattered from outer

segment to synapse, with endogenous mouse Prom1 and protocadherin 21 displaced in the same pattern. The authors ask that this be read with caution, because their western blots showed low PROM1 levels in those lines [2]. Conversely, severe recessive disease often suggests substantial functional deficiency, but biallelic inheritance alone does not prove that every allele is a complete biochemical null. Recessive missense alleles, transcript-specific effects and residual products remain relevant to treatment selection. [2,4–6,15,18]

Cehajic-Kapetanovic and colleagues provide a useful clinical and molecular foundation for this distinction. Their recessive series supports a deficiency-oriented therapeutic rationale, and says so: because homozygous missense and splice-site patients had phenotypes like those of similarly aged patients with truncating variants, the authors take that similarity to suggest that all recessive variants have a null, or close to null, outcome on PROM1 function. The clearest case is a woman in her twenties homozygous for the missense allele p.Leu618Arg, who had light perception only in both eyes and severe panretinal dystrophy. Nevertheless, incomplete parental phasing in the six of thirteen recessive patients who were compound heterozygous, and the absence of a direct functional assay for every variant, limit a universal molecular classification. The other seven were homozygous, so the phasing caveat does not apply to them; the absence of a functional assay applies to every variant in the series. Likely augmentation-responsive deficiency is a useful hypothesis. That all biallelic *PROM1* variants have been proven equivalent to a knockout is a stronger statement, and not one the sources assembled here advance. [5]

The Spanish study by Del Pozo-Valero and colleagues and the subsequent Italian series broaden the dominant–recessive phenotype literature beyond the frequently cited UK and East Asian reports. They are pertinent both to clinical classification and to the interpretation of unusual or mixed genotypes. The Spanish cohort screened 2,216 families with inherited retinal dystrophy and found *PROM1* variants in 32 of them, disease-causing in 18 recessive and four dominant families, with monoallelic or uncertain variants in the remaining ten; 25 patients from 22 families were phenotyped, most with cone–rod dystrophy and all with macular damage, and three of its variants are likely Spanish founder alleles. The Italian series reports a positional finding from its ten patients, nine male and one female: truncating variants in the N-terminal region were associated with earlier onset and greater functional impairment. Variant position and predicted consequence may help explain differences; in this ten-patient clinically ascertained series, the magnitude and generality of the association remain open. [60,61]

Two recent observations bear directly on that separation. Liu and colleagues' Chinese whole-genome study of 271 previously unresolved cases found three distinct *PROM1* deletions across four table entries, and one of them, a deletion of exon 15 at chr4:15992516–15997089 on GRCh38, appeared in both inheritance modes: heterozygous in three affected members of one family with a Stargardt-like phenotype, and homozygous or compound heterozygous in two further families with retinitis pigmentosa. A structural variant is therefore an allele class in its own right, and the same deletion did not behave the same way in every family. [89]

Kong and colleagues compared the two inheritance modes directly within one clinic, in a ten-person series. Four biallelic and six monoallelic individuals differed in mean reported onset, 9 years against 37.3 years, and in mean acuity, 20/200 against 20/30. Full-field electroretinography was performed in four biallelic and five monoallelic patients: two biallelic patients had extinguished photopic and scotopic responses, one had an extinguished scotopic and a severely attenuated photopic response, and the fourth, with cone dystrophy, had severely reduced photopic with preserved scotopic responses, while every monoallelic patient

retained normal scotopic responses and four of five had normal photopic responses. The small single-centre series indicates the direction of the difference; its size in a population remains open. A six-patient single-institution series by Lee, Abbey and Leys likewise reports more widespread recessive than dominant disease and identifies two truncating alleles, c.1909C>T and c.2050C>T. [90,113]

Within D’Esposito and colleagues’ ten-patient Italian series, an already reported father–son pair supplies a particularly informative mixed-genotype observation. The father carried c.1117C>T (p.Arg373Cys); the more severely affected son additionally carried maternally inherited c.1142-1G>A. The mother carrying the splice allele was clinically unaffected. This supports investigating interaction between a dominant allele and reduced function of the other allele, without identifying the interaction’s molecular mechanism. It is not a newly independent family for cohort pooling. [61]

The recessive account does not rest on recent series alone. Severe recessive retinitis pigmentosa was mapped to 4p15 and attributed to a *PROM1* mutation in a consanguineous Pakistani family in 2007; premature truncation was reported in autosomal recessive retinitis pigmentosa with early macular involvement in 2010; and a frameshift allele was described in cone–rod dystrophy in 2009. Taken together these establish that recessive *PROM1* disease spans retinitis pigmentosa with macular involvement and cone–rod presentations, that truncating alleles were among the first identified, and that the phenotype was recognised across separate populations well before the larger contemporary series. Liu and colleagues reported homozygous c.1445dupT in two unrelated Chinese patients with childhood-onset retinitis pigmentosa and non-recordable electroretinograms in young adulthood. In another Chinese family, Liang and colleagues found two affected brothers carrying a nonsense allele and a donor-site variant in trans; transfected-cell and minigene assays supported altered protein expression and splicing, respectively. A second dominant candidate allele has also been reported: c.734T>C, p.Leu245Pro, heterozygous across four generations of a Chinese family with Stargardt-like macular dystrophy, absent from 100 unrelated controls and segregating completely in the pedigree. The evidence offered for its mechanism is molecular-dynamics simulation rather than a functional assay, so it stands as a segregating candidate rather than as a second established dominant mechanism. [76–78,102,112,115]

3.2 Molecular diagnosis can revise the disease label

The dominant phenotypes arrived at *PROM1* by that route too. A kindred with an autosomal dominant Stargardt-like phenotype was mapped to chromosome 4p in 1999, with a maximum two-point LOD of 5.12 at zero recombination for D4S403 and a roughly twelve-centimorgan interval between D4S1582 and D4S2397, and the locus was named STGD4 [82]. A five-generation British family with bull’s-eye macular dystrophy apparent in the first or second decade was mapped in 2003 to 4p15.2–16.3, LOD 3.03 at zero recombination for D4S391, in an interval overlapping STGD4, and that locus was named MCDR2 [83]. Both names described a position on a chromosome rather than a gene, and both resolved to the same one. Two further phenotype aliases are in current use for the recessive presentations, CORD12 for cone–rod dystrophy and RP41 for retinitis pigmentosa, so one gene now carries four phenotype names across the two inheritance modes. When p.Arg373Cys was identified by positional cloning [2], it was present in every affected subject of the original STGD4 family and of the original MCDR2 family, and the 2010 series that set out the phenotype did so by revisiting those same pedigrees, which it labels families A and B [4]. Two locus names, four years apart,

resolved to one gene. A phenotype label records what was known when it was coined, which is the same reason the next case needed re-examination.

Martin-Gutierrez and colleagues' molecular reassessment of the British family historically labelled CORD7 changed its longstanding gene assignment. A known dominant *PROM1* p.Arg373Cys allele was identified in affected relatives, and an affected family member lacked the originally implicated RIMS1 variant. The reassignment rests on numbers. RIMS1 p.Arg820His was carried by ten individuals in gnomAD v2.1 of 108,049 and nine in v3.1.1 of 34,003, a frequency incompatible with a fully penetrant dominant cause; *PROM1* p.Arg373Cys was absent from gnomAD and had sixteen ClinVar submissions, fifteen pathogenic. Twenty-seven further carriers of the same *PROM1* variant were identified in the Moorfields Eye Hospital molecularly confirmed patient database rather than in ClinVar, and age-matched cases carrying it had phenotypes like the family's. A translocation between chromosomes 4 and 6 was tested for as an explanation of the original two-point LOD of 3.61 at 6q14 and was not found; Martin-Gutierrez and colleagues consider chance the likely explanation for the earlier linkage result. A LOD score is the base-10 logarithm of a likelihood ratio, so 3.61 means the observed segregation is about four thousand times more likely under linkage than under free recombination. The investigators concluded that *PROM1* explained the retinal disorder and that the case for RIMS1-related retinal dystrophy was weak. This is one family's diagnostic reassignment, reached by examining that family's variants. [59]

Noncoding disease is another important boundary of exon-only interpretation, and the three reported alleles do different things. One activates a pseudoexon deep in intron 18. A second, a deletion 17 to 26 bases from the intron 21 junction, causes in-frame skipping of exon 22 and a predicted 31-amino-acid deletion in the second extracellular loop, so it is neither missense nor truncating. What that source establishes is the predicted product: an in-frame loss of residues 761 to 791, a stretch the authors report as relatively conserved across 108 orthologues and containing two structurally important residues. Whether the shorter protein is made, is stable, or retains function was not measured, and the same paper reports that some exon 22 skipping occurs from the wild-type allele as well. It illustrates why predicted consequence and measured effect are different quantities. A third, c.221-20T>G, lies 20 bases from the exon 3 junction and was found homozygous in two Italian siblings whose clinical diagnosis had been Stargardt disease; it weakens rather than abolishes the acceptor consensus, and it fell outside a targeted panel's reporting filters on distance and prediction score. [65,73,88]

The route to the pseudoexon allele shows what exon-focused sequencing does not reach. Mayer and colleagues studied a consanguineous family with two affected siblings. Homozygosity mapping returned an interval containing two genes already associated with recessive cone-rod dystrophy, *RAB28* and *PROM1*, and sequencing the exons and canonical splice sites of both returned nothing. Whole-genome sequencing found 18 single-nucleotide variants in *RAB28* and 196 in *PROM1* shared by the siblings; after annotated variants were filtered out, one private variant remained, c.2077-521A>G, deep in intron 18, segregating in the expected recessive mode. Minigene splicing reporters then showed a pseudoexon integrated into the mutant transcript, giving a premature termination codon and a presumed functional null. The same allele appears independently in an American family, as one of two compound heterozygous alleles, alongside a 7,340-base deletion of two exons in a second family that whole-exome sequencing had left with a single heterozygous variant. [65,93]

These reports motivate transcript-aware interpretation of noncoding variants, and whether patient RNA can supply it depends on the isoform and the region. For the intron 21 deletion,

reverse-transcription polymerase chain reaction (RT-PCR) for *PROM1* expression in white blood cells and skin fibroblasts was negative, so that allele was assigned by minigene assay; those are among the accessible tissues RNA diagnostics uses in other genes. The long isoform NM_006017.3 is expressed at very low levels in blood, but exon 3 is present in every isoform, and the shorter NM_001145847.1 amplified from whole-blood complementary DNA well enough to show complete exon 3 skipping in a patient and none in a control, at 579 against 635 bases. Two of the three published noncoding alleles were therefore assigned by minigene assay and the third by patient RNA. Patient RNA can resolve a noncoding *PROM1* candidate where an accessible isoform carries the affected exon, and functional assay and segregation can resolve it where no accessible isoform does. These routes provide the allele-specific characterisation on which splice-directed therapeutic experiments depend, and a result for one allele does not establish that the same splice repair will work across unrelated variants. [65,73,88]

Tracewska and colleagues' Polish national cohort shows the same revision happening three times in one dataset. Among 190 families screened on a 108-gene panel, 115 were solved, and the report states that diagnosis was re-evaluated in 16 cases. Its three *PROM1* families all ended with the same final label, retinitis pigmentosa with macular involvement, from three different starting labels: Stargardt disease in a woman homozygous for c.1697dupA, p.(Asn566LysfsTer2), with onset at three and diagnosis at seven; bull's-eye maculopathy, Stargardt disease or retinitis pigmentosa in a woman homozygous for c.1612delT, p.(Trp538GlyfsTer15); and retinitis pigmentosa, cone or cone-rod dystrophy in a man compound heterozygous for c.439C>T, p.(Gln147Ter), and c.1142-1G>A, the same splice allele carried by the Italian mother described above and here in an unrelated family. A fourth patient, with achromatopsia and a single heterozygous CNGB3 frameshift, carried one *PROM1* c.1697dupA allele that the authors record as a possible contributor rather than a second diagnosis. Khan and Bolz reported the same c.1697dupA allele online in February 2014 in a Saudi Arabian woman with childhood-onset cone-rod disease, high myopia and nystagmus. [87,114]

Puertas-Neyra and colleagues' c.1354dupT study in induced pluripotent stem cells (iPSCs) informs experimental design: different clinical presentations occurred among individuals sharing the reported variant, and correction restored a cellular molecular readout. Three patient backgrounds and two corrected lines provide genetically informative cellular comparators; independent demonstration of restored human retinal function remains open. [12]

Two 2026 cohorts put numbers on how often a diagnosis is revised this way. Wang and colleagues reinterpreted the sequencing results of 246 patients carrying a clinical diagnosis of Stargardt disease at one United States referral centre. Laboratory reports classified 195 of them as positive or likely positive for any inherited-retinal-disease gene, 173 and 22 respectively: 191 ABCA4, three PRPH2 and one *PROM1*. Phenotype-integrated reinterpretation then raised the ABCA4 positive or likely positive count from 191 to 224, and the authors describe both PRPH2 and *PROM1* as well-described phenocopies of ABCA4 disease. Harford and colleagues came at it from the other side. Of 232 patients with macular or cone dystrophy, 72 were still unsolved after first-line testing, and second-line sequencing resolved 35 of those: three of the 72 carried heterozygous *PROM1* variants, at a mean age of 55 with a range of 50 to 63. One number describes how often a clinical Stargardt diagnosis turns out to be *PROM1*; the other, how often *PROM1* is waiting behind a negative panel in later life. [107,108]

These reports have a practical consequence for people already diagnosed. Where a clinical diagnosis of Stargardt disease, bull's-eye maculopathy, cone-rod dystrophy or retinitis pigmentosa sits alongside a negative panel, or, where biallelic disease is suspected, alongside

one *PROM1* allele with no second one found, the cases above show what can be missed: a pseudoexon deep in an intron, a variant a targeted panel had already seen and filtered out on distance and prediction score, and a deletion spanning two exons. Each was reached by genome sequencing or by patient RNA rather than by re-reading the exons, and one national cohort re-evaluated sixteen diagnoses in a single dataset. Whether re-testing is warranted for a particular person is a question for their genetics service; the point here is only that a negative or single-allele exon-based result is not a closed one. [65,87,88,93]

3.3 International evidence and denominators

Japanese, Korean, Chinese, Taiwanese and Indian studies prevent a geographically narrow picture of *PROM1* disease. Their sampling frames are nevertheless different: a *PROM1*-focused clinical series, a Stargardt-ascertained cohort and a broad inherited-retinal-disease sequencing cohort cannot supply interchangeable prevalence denominators. The Japanese report included ten patients from eight families and a mixed set of variant classifications; the Korean report included seven patients from five families. Each discloses one patient whose data had already appeared elsewhere: the Japanese series names a 2016 paper by the same group, and the Korean series names Kim and colleagues' 2017 report of one patient with p.Arg373Cys. Both overlaps are small, but a reader adding the two cohorts together is double-counting those individuals if the two earlier reports are counted as well. In the supplement to Suga and colleagues' Japanese sequencing cohort, five *PROM1* patients belonged to four pedigrees; four patients are attributed there to Fujinami and colleagues' previously reported series. A 2025 Shanghai conference abstract reports 33 patients with *PROM1* variants, 16 classified as dominant and 17 as recessive, with earlier and more widespread disease in the latter group. A reported variant is not thereby an established pathogenic allele, and a publication is not thereby an independent family. [6,13,15–18,116–118]

A further 40 multi-gene cohorts tested fewer than 100 people each and are not tabulated. Thirty-five of them report at least one *PROM1* patient or family and five report none. At that size, neither outcome gives a stable estimate on its own. The five reporting none tested 194 people or families between them, where about two *PROM1* cases would be expected at the share of about 1% that most of the larger cohorts show, and several of the 35 report a single *PROM1* family among 20 or fewer tested, a share of 5% or more and several times what most of the larger cohorts report. Their contribution is that *PROM1* has been sought across a wider set of populations than the tabulated cohorts alone would suggest, not a frequency in any of them. Each of the 40 is listed, with its size and its *PROM1* figure in the unit it reports, in the data file described under Data and materials.

One registry has published the dominant and recessive split directly. Francisco and colleagues' Portuguese cross-sectional study searched 40 genes reported with dual inheritance, found 22 in its registry and nine displaying both patterns across 102 families and 141 patients; within *PROM1*, 76.9% of cases were recessive, spanning retinitis pigmentosa, macular dystrophy and cone-rod dystrophy. Dual-inheritance genes accounted for 12% of that registry's genetic diagnoses. This is a referral-centre proportion from one country rather than a population estimate, and it rests here on the published abstract. [95]

The large UK molecularly characterised cohort is useful for the contribution of genes within that service population, not population prevalence. Its *PROM1* entry records inheritance as "recessive and dominant" and reports 37 families and 53 affected individuals, each 1.2% of the cohort. Because that single figure is not resolved by inheritance mode, it supports neither

a dominant nor a recessive frequency estimate, and it cannot be repurposed as one. The Taiwanese series reports two sets of unaffected heterozygous carriers and reads them two ways. Those carriers hold the frameshift alleles c.139del and c.794del, which is the expected carrier state in recessive disease; the authors also take the c.139del carriers as evidence of incomplete penetrance of dominant *PROM1* alleles. That second reading concerns dominant variants as a class, not p.Arg373Cys in particular, and it comes from a small ascertained series without an estimated penetrance. Ascertainment, inheritance and allele identity are part of what a denominator means, and a figure separated from them is hard to interpret. [14,18]

3.4 Modifier biology and genetic background

Lee and colleagues described a family in which a *PROM1*-associated phenotype differed in the presence of an *ABCA4* variant. The observation supports a human interaction hypothesis; controlled backgrounds and biochemical phenotypes are needed to distinguish a trans-modifier from additional genetic background, a second diagnosis and other sources of variable expression. The reported family carried *PROM1* p.Arg134Cys rather than p.Arg373Cys, so the observation does not transfer automatically to the allele that dominates this literature. A factorial genotype comparison on controlled backgrounds, accompanied by a biochemical phenotype, could distinguish these possibilities. [56]

This caution has a positive consequence for research: patients with the same *PROM1* allele but different trajectories are useful sources of modifier hypotheses. Isogenic correction tests the contribution of *PROM1* within a donor background, while reverse knock-in and multiple independent donors test whether the result transfers across backgrounds. Neither strategy alone establishes a population-level modifier effect. Variant interpretation, patient-derived modelling and natural history therefore develop together as mutually informing lines of evidence.

4. From membrane architecture to renewal and clearance

4.1 *PROM1* localisation, membrane organisation and signalling

Prominin-1 is a pentaspan membrane glycoprotein that concentrates in membrane protrusions rather than in flat membrane, and in the photoreceptor it sits at the base of the outer segment, where new disk membrane is made. What it does there is not settled. The evidence places it in the organisation of outer-segment membrane rather than in the formation of individual disks, because disks still evaginate, fuse and hold their spacing without it. Beyond that, insufficient functional protein, abnormal stoichiometry with membrane partners, altered lipid association and defective processing all remain live candidates, and the experiments reviewed below separate them only in part. In *Xenopus* photoreceptors, Han and colleagues found prominin-1 at open cone-lamellar and basal rod-disk rims, opposite peripherin-2 at closed rims. Comparative retinal mapping also found prominin-1-positive inner nuclear layer populations in zebrafish, chicken and mouse, with species-dependent distribution. [84,86,119,120]

Loss-of-function and dominant-mutant studies place *PROM1* in the organisation of photoreceptor outer-segment membranes. However, an image showing altered localisation does not by itself distinguish a primary membrane-shaping defect from defective trafficking, altered stability or secondary injury. The relevant causal chain includes transcript abundance, glycosylation, membrane delivery, lipid interaction and the geometry of nascent disks. Different experiments interrogate different links in that chain. [2,3,7,9]

Carr and colleagues tested that placement directly in *Xenopus laevis*. CRISPR-generated *prom1*-null animals had severely dysmorphic photoreceptors with overgrown and disorganised disk membranes, and cone outer segments were affected more than rods. Disk membrane evagination, disk fusion and the spacing between disks and lamellae all persisted, which places the requirement at the higher-order organisation of the outer segment rather than at disk formation. The same study generated *cdhr1*-null and *prom1* plus *cdhr1* double-null animals: *cdhr1*-null photoreceptors were not grossly dysmorphic, although some disks were overgrown or vertically oriented within the plasma membrane, and double-null animals did not differ significantly from *prom1*-null animals. The authors read the difference in severity between the two single nulls as distinct roles in disk morphogenesis rather than one shared pathway. A two-gene test in one species leaves the corresponding question for the human proteins open. [86]

A zebrafish line offers a candidate reason for the cone asymmetry. Loss of *prom1b*, the orthologue that matters in zebrafish photoreceptors rather than *prom1a*, disrupted outer-segment morphogenesis with the same direction of effect: cones degenerated early, while rods stayed viable with abnormal outer segments as late as nine months after fertilisation. In wild-type fish the authors located Prph2, the peripherin 2 orthologue, at the edge of the outer segment and found it more abundant in cones than in rods, and *prom1b* deletion both mislocalised Prph2 and disrupted its oligomerisation. Their proposal is that the difference in Prph2 between the two cell types is one reason they fail differently. The mouse knockout does not reproduce it. There, peripherin 2 was less abundant on immunoblot but showed no evident displacement at the ages examined, so the reported localisation phenotype differs between the two species. That distinguishes the two results without establishing that the mouse defect is confined to abundance, since not every functional property of the protein was tested, or that either the cone selectivity or the mechanism proposed in the fish is shared. [94,105]

Outside the retina, Lee and colleagues report a signalling role for prominin-1 that ends at cholesterol. In mouse sensory neurons, deleting *Prom1* inhibited axon regeneration after sciatic nerve injury, and raising prominin-1 by viral delivery enhanced regrowth, including on an inhibitory substrate. The mechanism offered is that prominin-1 binds the type I transforming growth factor beta receptor ALK4 and, with it, drives phosphorylation of Smad2, which lowers the expression of cholesterol-synthesis genes and reduces cellular cholesterol. The cholesterol effect was measured in cultured dorsal root ganglion neurons, and its direction has not been tested in either retinal compartment. What it adds is that prominin-1 sits in a receptor complex with a transcriptional output, which the retinal experiments have not examined. [98]

Bell and colleagues add a direct biophysical line of evidence. In cultured mammalian cells, human *PROM1* and TTYH1 promoted extracellular-vesicle formation; their vesicles differed in membrane tubulation and cholesterol interaction. The retinal-disease-associated mutant tested was W795R, a transmembrane-domain substitution, which showed particularly stable cholesterol binding. No equivalent membrane-biophysical measurement has been reported for p.Arg373Cys, so this line of evidence does not yet reach the common dominant allele. These experiments connect PROM1 to membrane bending and lipid-dependent vesicle behaviour, which colocalisation with an outer-segment marker does not show. Their relevance to retinal disease is mechanistic: they measure membrane behaviour in cultured cells, several steps upstream of vision. In cultured RPE1 cells, Hori and colleagues found that overexpressed prominin-1 produced cholesterol-enriched membrane fibres requiring Rho/ROCK activity; in mouse embryonic fibroblasts, *Prom1* loss altered calcium-stimulated chloride efflux. [55,121]

The synthesis is therefore a membrane-homeostasis model with several potentially separable failure modes. Insufficient functional PROM1, abnormal stoichiometry with membrane partners, excessive or altered lipid association and defective processing could converge on outer-segment disorganisation. Rescue by a particular construct tests the combined consequences of these processes. It does not automatically identify which one was primary.

Xiao and colleagues measured what happens to the other outer-segment proteins. In a CRISPR *Prom1* knockout on a C57BL/6 background, semiquantitative immunoblots at ten days of age, before any evident degeneration, found peripherin 2, Rom1, Pde6 β , rhodopsin and protocadherin-21 reduced by 22% to 47%. The later pattern is protein-specific rather than uniform: Gnat1 and Grk1 were at or near wild-type levels at ten days and significantly reduced by four weeks, while Rom1 and rhodopsin were no longer significantly reduced at four weeks. For Gnat1 and Grk1 this is the timing given in the source's tables and discussion. Delivery failed separately from abundance, and selectively: rhodopsin was displaced into the outer and inner nuclear layers at two weeks but not at six, cone opsin was normally placed at two weeks and displaced into the outer nuclear and outer plexiform layers at six, and peripherin 2, Grk1 and Pde6 β showed no evident displacement at eight or twenty-four weeks. Outer nuclear layer thickness fell 47% by sixteen weeks and 84% by twenty-four on optical coherence tomography, and scotopic electroretinogram a-waves fell 67% at eight weeks and 88% at sixteen, with four animals per group. A protein can therefore be scarcer without being misplaced, and neither measurement on its own says whether less was made, less was retained or less arrived. [105]

4.2 Isoform, abundance, epitope and function are separable quantities

Human retinal splicing is developmentally regulated. Moya-Molina and colleagues studied a retina-associated *PROM1* exon and linked its perturbation to photoreceptor maturation and cellular phenotypes in human models. The reported approximately 57% exon skipping by RT-PCR and approximately 55% reduction in exon-containing RNA by qPCR are different assay outputs, not two estimates of an identical quantity. Reduced frequency of ciliated cells is a different measurement from shorter cilia. [7]

A separate mouse literature concerns exon-dependent recognition by the 13A4 antibody. Matalkah and colleagues demonstrated a structural epitope, and quantified its effect on measurement: in wild-type mouse retina the antibody reported approximately three to five times less PROM1 than a second antibody from postnatal day 13 onwards, and placed the developmental peak five days early [19]. The signal is not lost so much as differentially weighted, because the antibody still recognises both isoforms with reduced affinity for one. Loss of antibody signal therefore does not by itself establish loss of all PROM1 protein. The exon behind this mouse epitope problem is also not the human retinal exon, and the mapping between them is known: the mouse microexon is exon 19, eighteen nucleotides, six amino acids, in the third extracellular domain, which is the second of the two large loops, and it is not used in primates including humans because mutations disrupt its splice sites; the human retinal exon is exon 4, twenty-seven nucleotides, nine amino acids, predicted to sit on the extracellular side close to the first transmembrane helix, which is established by the splicing study rather than by the antibody work [7]. They are different exons, of different length and in different domains, and the mouse one is present in the human genome but spliced out of the normal human transcript. When the human gene was linked to retinal disease, mouse prominin and human PROML1 were reported at an average of about 60% amino acid identity. In MDCK cells, Jászai and colleagues linked engineered prominin-1 splice variants s1 and s2, carrying the substitutions Y819F and Y828F, to shorter cilia and impaired association with Arl13b; zebrafish

prom3 knockdown also altered Kupffer's-vesicle cilia and left-right patterning. These findings connect comparative prominin biology to ciliary function, while their transfer to human retinal alleles remains open. [20]

Two experimental questions are distinct. An equal-input comparison estimates the total effect of delivering two constructs, including any effects on RNA stability, translation, glycosylation and trafficking. An abundance-matched comparison asks whether the proteins differ at a selected expression level. Both are useful, but the second may remove a clinically important mechanism of the first. Apical localisation and cell maturation are possible mediators, and matching them away removes what they might explain. Orthogonal antibodies, transcript assays and protein measurements can then distinguish epitope loss from altered abundance and altered function.

4.3 RPE autophagy: the functional flux evidence

The RPE evidence includes functional assays, not only static markers. Alongside the 2017 human-RPE study, Bhattacharya, Yin, Huo and Chaum used HPV E6/E7-immortalised mouse RPE with CRISPR-mediated *Prom1* loss. LC3-II and p62 responses to bafilomycin supported reduced autophagic flux; p62 transcription also increased. The study measured mTORC1/TFEB signalling and nuclear localisation, and cathepsin-D maturation by immunoblot, not a separate substrate-turnover activity assay. Rhodopsin immunoblots after continuous bovine outer-segment feeding supported altered clearance. Binding, ingestion and digestion are not separated by that design, but these results support an RPE mechanism without establishing that it initiates every human *PROM1* retinopathy. [21,53]

By contrast, a change in LC3 or p62 abundance at one time point cannot by itself determine flux. Increased LC3-II can reflect increased autophagosome production, slower degradation or both; reduced abundance can also have several explanations. Consequently, the c.2321delC report and static measurements in other models do not resolve into a simple "autophagy increases" versus "autophagy decreases" contradiction. Allele, tissue, maturation, stimulus and assay differ across these studies, and results are comparable only where they align. [7,22,53]

Clearance is itself a sequence of processes. Outer-segment binding and engulfment, phagosome maturation, lysosomal digestion, macroautophagic flux and LC3-associated phagocytosis are not interchangeable endpoints. An assay that reports persistence of an ingested outer-segment label may identify a clearance defect without locating it to a single step. Pulse-chase cargo assays, lysosomal function and validated flux measurements with targeted rescue could distinguish these steps, because a single autophagy label does not identify which of them failed.

4.4 RPE injury in vivo: localisation and initiation

Bhattacharya and colleagues' 2024 study used subretinal delivery of CRISPR components in an adeno-associated virus (AAV) serotype 2/1 vector to reduce *Prom1* in local RPE regions. The report describes regional RPE degeneration, associated photoreceptor abnormalities and reduced ERG responses, with no detected reduction of photoreceptor *Prom1* expression in the sampled analysis. Scrambled-guide controls and spatial expression measurements strengthen the link between local RPE perturbation and retinal damage, which a whole-animal knockout cannot localise by cell type. [10]

A regional, RPE-biased perturbation supports an RPE contribution while leaving cell-type-specific causation, and the initiating lesion across human alleles, open. Editing efficiency, transduced-cell distribution, local injury and subsequent propagation remain important. The study reported no evidence of an inflammatory response attributable to delivery, and the

control eyes, injected with the same vector carrying a scrambled guide, kept a normal fundus. Reciprocal photoreceptor- and RPE-selective perturbations with temporal and spatial mapping could separate the contributions of the two compartments. [10]

In prominin-1-null *Xenopus*, subretinal drusenoid-like deposits, cone–rod dystrophy and RPE atrophy broaden the anatomical phenotype beyond a rod-dominated mouse readout. They reinforce the importance of the photoreceptor–RPE interface. Because the mutation affects both compartments, locating cell autonomy and the site of initiation needs an experiment that separates them. [24]

The connection with atrophic age-related macular degeneration (AMD) is mechanistic as well as anatomical. Bhattacharya and colleagues place *PROM1*-associated disease within a broader account of RPE autophagy, lysosomal clearance and epithelial homeostasis. In disrupted mouse RPE, bafilomycin-supported flux changes and altered outer-segment clearance connect deficient *PROM1* function to processes implicated in AMD. Regional RPE-biased perturbation in mice and subretinal drusenoid-like deposits with RPE atrophy in prominin-1-null *Xenopus* provide complementary *in vivo* evidence. These findings support the use of *PROM1*-deficient systems to investigate aspects of atrophic AMD biology. [10,11,21,24,53]

The resemblance does not by itself establish that *PROM1* variants confer human AMD risk or that the initiating lesion is always in the RPE. In the frog, outer-segment dysmorphology and shedding abnormalities coexist with later RPE injury; the whole-animal mutation cannot separate their contributions. The mouse regional perturbation localises an RPE contribution without providing an allele-matched comparison of initiation in the two compartments. A shared-pathway hypothesis predicts that correcting the same clearance defect improves both *PROM1*-deficient and AMD-relevant RPE phenotypes at comparable stages. Differences in maturation, cellular ageing, genotype and injury type could account for discordant responses; rescue confined to the deficient system would restrict the proposed therapeutic transfer. Neither model demonstrates an AMD treatment effect in people. [10,11,24,53]

4.5 Transcriptomic convergence and pathway-specific responses

Huo and colleagues profiled cultured mouse RPE after *Prom1* disruption and linked the changes to stress, inflammatory signalling, altered epithelial homeostasis and clearance-related genes. The strong separation shown on a principal component analysis (PCA) plot is in the batch-adjusted analysis set, which excludes two knockout samples. [27]

The paper also contains an important biological nuance: the Hallmark epithelial–mesenchymal transition (EMT) signature was reduced, while selected EMT-associated genes and validation experiments supported a partial or context-specific change. These are not equivalent claims. Huo and colleagues themselves interpret the pattern as a stalled or incomplete EMT, in which the cells begin the transition without engaging the complete mesenchymal programme; its features are gene- and clone-dependent. Similarly, pathway enrichment and expression of a mitophagy-related gene are leads to functional experiments, not direct measurements of pathway flux. [27]

The available dataset identifiers, including GSE298665 for this RPE study and DRA016296 for the acute-light single-cell study, are starting points for reanalysis. Sample metadata, clone identity, exclusions and biological replication all bear on a pathway-level estimate. [26,27]

5. Model identity, light and the initiation–propagation distinction

5.1 Knockout lines are not interchangeable

Table 2 lists the experimental models drawn on here, what each is used for and how far each transfers. Dellett and colleagues compared a C57BL/6 *Prom1*-null background with a hybrid C57BL/6×CBA/NSIc background and found substantially different degeneration trajectories. Histology compared the backgrounds, whereas the remaining experiments used the hybrid line. A result from a *Prom1* mouse therefore depends on four things besides the gene itself: the strain's genetic background, how the mutation was built, whether any prominin-1 protein is still made, and how much light the animals received. Two studies that differ in any of them can disagree for that reason alone. Table 3 sets out, for each mouse study, what its source reports on all four. [25]

Yanardag and colleagues measured reduced photoreceptor responses from postnatal day 12, before eye opening and before any detectable loss of photoreceptors. That timing bounds how far a light effect reaches: a structural defect can precede overt degeneration or normal eye opening, while light modifies the injury that follows. Dark-rearing protection in one line does not establish that another line, allele or human disease is entirely light-dependent. The remaining question is how baseline structural abnormality, photic stress and progression interact. [23,25]

The *Xenopus* nulls draw the same separation in another species. Outer-segment disorganisation there was severe while photoreceptor loss was neither early nor severe. The scotopic electroretinogram was affected little, with a reduced A-wave at the lowest intensities and no significant B-wave change, and the photopic response was significantly reduced at higher intensities. Neither response was abolished, and the F1 repeat matched the F0 result apart from a greater depression of the scotopic B-wave. The authors read this as evidence that disk disorganisation is not itself what kills the photoreceptor, and that an indirect secondary effect, such as the deposits that accumulate in the outer-segment layer, carries the degeneration. The sequence was proposed, not measured directly, and its transfer from a non-mammalian retina remains open. [86]

Similarly, the classic dominant p.Arg373Cys mouse work used mutant expression in a transgenic setting; an endogenous heterozygous knock-in would answer a different question. Its evidence for disruption is important, but expression level, stoichiometry and insertion context matter when planning a therapy intended for a human endogenous allele. Existing transgenic and patient-derived resources are distinct from the proposed new knock-in lines. [2,5,12]

5.2 Acute light challenge and chronic exposure answer different questions

The Shigesada single-cell discovery experiment used two male *Prom1*-knockout littermates: one received an acute light challenge after surgical opening of the eyelids and pupil dilation, and one remained dark-reared. The exposure was 15,000 lux for three hours at postnatal day 11, followed by overnight recovery. Thus, there was one animal per condition for that discovery comparison. Thousands of recovered cells increase cellular resolution but do not estimate between-animal variation. [26]

The paper also reports separate histological, molecular and treatment experiments. The one-animal-per-condition design constrains the evidential role of the discovery dataset: cell-type-localised candidates, subsequently tested by other experiments, rather than a biologically replicated population estimate of the acute transcriptional effect. That endogenous

IGF1 protein was not quantified directly is a separate matter from the pS6 measurements and the response to supplied IGF1. [26]

Lux alone does not describe an exposure. Spectrum, duration, retinal irradiance, pupil and lens state, circadian phase, prior adaptation, cage geometry and temperature affect interpretation. Surgical opening, acute challenge and ordinary cyclic illumination are three different exposures. These distinctions support controlled light-modification experiments; the mouse dark-rearing protocol alone does not establish a human preservation effect.

6. Interventions: what has been rescued?

6.1 *PROM1* replacement in deficient systems

Ji and colleagues delivered a reported 2,598-base wild-type human *PROM1* coding sequence under a short rhodopsin-kinase promoter in a serotype 2 mutant capsid, by subretinal injection into *Prom1*-knockout mice. Their methods cite the genomic reference NG_011696.2, which locates the gene rather than identifying the transcript isoform delivered. The working dose was one microlitre of 1×10^{12} vector genomes per millilitre at postnatal day 14, which is 1×10^9 genomes in total. Retinal protein expression did not differ significantly between that dose and a five-fold higher one, which is an expression readout and not a demonstration of equivalent rescue. Every animal's fellow eye received buffer, so the comparison is paired within an animal rather than between groups.

Separate arms answer the timing question. Two further cohorts were treated at postnatal day 21 or 28, after the outer nuclear layer had already thinned by about 23% and 35% against wild type, and each received twice the volume, 2 microlitres, so twice the total dose. Both were read out at the reported five-month endpoint, which the paper describes once as five months after injection and once as five months of age. In the day-21 cohort the treated nasal side ran from 32.3 ± 5.7 to 34.9 ± 4.4 μm against 16.6 ± 1.5 to 20.6 ± 2.0 μm in the vehicle-treated fellow retina, and in the day-28 cohort from 29.2 ± 2.4 to 36.8 ± 5.5 against 14.6 ± 2.6 to 19.8 ± 1.3 . These are ranges of location-specific means rather than of individual animals. Dark-adapted electroretinogram responses were better preserved in the day-21 cohort, on five paired comparisons, and expression there was widespread across both the injected nasal side and the temporal side of the same retina.

Two qualifications are the authors' own, and both belong to the day-14 cohort, which was followed at two, three and five months. No significant difference in dark-adapted electroretinogram amplitudes appeared at three months despite comparable protein expression at three and five months, which the authors read as functional recovery needing longer than expression does; and the treatment did not prevent the slow deterioration of amplitudes over time. So benefit is partial rather than restorative and can be slow to appear, and the effective window extends at least to an age at which roughly two thirds of the outer nuclear layer remains. How much viable tissue is required, and how far the benefit spreads from the injection bleb, remain open. The authors declare the commercial interest behind the work: Fenghua Wang owns Innostellar Biotherapeutics, Haiyan Ji and Zhuoyu Ni are its employees, and both the coding cassette and the capsid are the subject of patent applications. Section 6.2 sets out how those filings and this paper relate to one another. [8,28,29]

A rhodopsin-kinase promoter is used by three of the constructs here: Ji and colleagues' delivery work above, Holmes and colleagues' knockout-mouse programme below, which is reported at meeting-abstract level, and the 2024 patent filing of Section 6.2, which pairs that

promoter with a wild-type coding sequence. Xu and colleagues' organoid and mouse work, which follows, uses a CRX promoter instead. The rhodopsin-kinase promoter has been validated independently, which is why its cell-type coverage can be reasoned about even where a construct's own methods are not available, and that validation does not carry across to CRX. A rhodopsin-kinase/GRK1 promoter has been shown to drive expression in cones as well as rods. The derivative tested was the human sequence from -112 to +180, carrying the full 5' untranslated region, packaged in an AAV2/5 vector and injected subretinally in wild-type mice, in the all-cone *Nrl*-null mouse and in the cone-rich diurnal Nile grass rat. Control eyes in the same experiment received a similar vector carrying a mouse rod opsin promoter. In rods the two promoters were comparably efficient, with the rhodopsin-kinase promoter reported as the more uniform; in cones the rhodopsin-kinase promoter drove expression and the opsin promoter did not. Expression was detectable at two weeks and stronger and more widespread at six to eight weeks, covering nearly the full expanse of the retina, and it remained so at sixteen [80]. In nonhuman primate retina, an AAV5 vector carrying the same promoter and driving green fluorescent protein (GFP) was delivered subretinally to three male *Macaca nemestrina* aged six to seven years, and immunohistochemistry found the promoter active in and specific for both rods and cones. Cones in the rod-free foveal pit were examined as well, and the authors report that the vast majority of photoreceptors there carried the reporter; the two sections selected for analysis were parafoveal, taken 550 and 850 micrometres superior to the foveal pit [81]. Cone expression is therefore the expected default for this promoter, and rod-exclusive expression would need direct demonstration in the particular vector, promoter derivative and species used. [80,81]

Xu and colleagues' final published abstract identifies a homozygous c.619G>T (p.Glu207Ter) patient model with transcript decay, altered splicing and absent PROM1 protein. AAV7m8-CRXp-hPROM1 rescued outer-segment-like structures in patient retinal organoids; AAV8-CRXp-hPROM1 preserved outer-segment morphology and improved visual function in *Prom1*-null mice. These vectors establish rescue in deficient organoid and knockout-mouse models; dominant-allele rescue remains open. Photoreceptor-associated splicing and N-glycosylation strengthen the rationale for processing-aware cargo selection. Detailed dosing and treatment-window comparisons are unavailable in the final record consulted. The publisher's indexed data-availability statement identifies RNA-sequencing accession GSE299857; the underlying data were not reanalysed here. The induced pluripotent stem cell lines behind that patient model were described separately: an affected homozygote and an unaffected heterozygous mother from one family, both carrying c.619G>T, p.Glu207Ter, characterised for pluripotency and genomic stability without retinal differentiation or rescue. That report supplies the lines' family provenance and validation, not a retinal result. [9,101]

A third programme, from Holmes and colleagues, has been reported at meeting-abstract level, and it speaks to the window question directly. Adeno-associated virus vectors carrying human *PROM1* under the human GRK1 promoter, packaged in AAV2/8 Y733F capsids, were injected subretinally into *Prom1*-knockout (Rd19) mice at postnatal day 15, during early degeneration, and at day 30, after the reported loss of about half the photoreceptors. The day-30 group is reported with rescue of both scotopic and photopic electroretinogram responses, and the day-15 group with long-term recovery of scotopic function from one to ten months after injection, and the authors report outer-segment architecture on transmission electron microscopy, restored outer-segment protein localisation and preservation of photoreceptor numbers against saline-injected eyes. At abstract level, the report supports qualitative rescue after day-30 treatment and durable scotopic recovery after day-15 treatment;

vector-concentration exponent, animal counts, statistical treatment and spatial coverage are unavailable in the record consulted. Its bearing is that a second laboratory reports benefit from delivery after substantial photoreceptor loss, in a knockout line, with a promoter whose cone activity is documented separately. [103]

For dominant disease, augmentation remains a testable strategy. Increased wild-type protein might overcome a stoichiometric deficit or partly compensate for a disruptive product. Conversely, excess expression could fail to remove a toxic interaction or itself disturb membrane organisation. A dose series in an endogenous heterozygous mutant, compared with a heterozygous null and a corrected control, is what separates these possibilities; the inheritance label alone does not. [2,5,8,9]

6.2 Splice repair, correction and construct disclosures

Correction of c.1354dupT in patient-derived iPSCs and experimental manipulation of a retinal *PROM1* exon show that transcript- and allele-specific resources are feasible. The three noncoding reports differ: one is a deep intronic variant activating a pseudoexon, one a deletion close to the intron–exon junction causing in-frame exon skipping, and one a non-canonical splice-site variant whose effect was read directly from patient RNA. Each gives an additional rationale for splice-directed intervention. These are distinct starting points: repairing a DNA sequence, correcting an abnormal splice event and supplying a replacement transcript have different delivery and expression requirements. Molecular correction and visual rescue are different endpoints, and mature retinal phenotypes lie between them. [7,12,65,73,88]

Two Chinese patent disclosures set out *PROM1* vectors, and they are separate filings rather than members of one patent family. The earlier claims a codon-optimised *PROM1* coding sequence in a cassette of CMV enhancer, chicken β -actin promoter, SV40 intron and SV40 polyA, packaged in adeno-associated virus. The later uses a short rhodopsin-kinase (RK1) promoter with a wild-type coding sequence, and its reported comparison sets that construct against the earlier one. Promoter and coding sequence differ together in that comparison, so it does not separate the effect of the promoter from the effect of codon optimisation, and a promoter restricted to photoreceptors set against one active broadly also changes which cells contribute to any measured signal. The published study of Section 6.1 settles which of the two its construct resembles: it uses a wild-type coding sequence under a short rhodopsin-kinase promoter, which is the later cassette rather than the codon-optimised earlier one. It cites the later filing directly, by application number 202410700899.1, which had not published when the article was written: it was filed on 31 May 2024 and published as CN121046460A on 2 December 2025, after the article was accepted. A separate application covers the capsid. The published experiment and the later disclosure are therefore documented as the same programme rather than inferred to be, although a shared application number establishes neither identity of every cassette version nor that the patent and journal experiments are the same experiment. A single group is confirmed directly. Shanghai Langsheng Biotechnology, which holds the filings, is the first affiliation of three of the five authors of the published paper, whose declared commercial interest is in Innostellar Biotherapeutics; it is also the institution whose animal committee approved the work. These filings are read here as construct and development disclosures. Sequence length, translation and motif claims require the actual sequence, version and reproducible computation. [8,28,29]

Three cited patent filings come from the same company as the gene-therapy paper, and together with that paper they are one programme rather than separate lines of evidence.

All three are assigned to Shanghai Langsheng Biotechnology, and their inventor lists give Sun Xiaodong on the 2018 filing and Ji Haiyan, Ni Zhuoyu and Wang Fenghua on the two filed in May 2024, which correspond to four of the five authors of the gene-therapy report cited above. They are therefore counted here as one line of development. [8,28,29,104]

The disclosure texts and the published paper carry construct detail independently. The earlier filing differs from the later one in both the promoter and the coding sequence, and the published construct agrees with the later filing on both, at the level each states: a short rhodopsin-kinase promoter and a wild-type coding sequence. Agreement at that level is not identity of the whole cassette. The 2018 filing reports its optimised coding sequence expressing about 3.2 times the unoptimised sequence on immunoblot in human embryonic kidney (HEK-293) cells, and electroretinography in *Prom1*-knockout mice injected at postnatal day 14, where the scotopic a-wave of eyes receiving the optimised vector was not significantly different from normal controls while the unoptimised vector performed less well. The 2024 filing states that its coding sequence was amplified from human retinal organoid RNA, pairs the RK1 promoter with an IRBP enhancer, reports expression confined to the photoreceptor layer with a distribution the filing describes as matching wild type, and gives an effective dose range spanning seven orders of magnitude, which is a claim boundary rather than a dose. A third filing from the same company discloses the human model used alongside them: a guide RNA targeting *PROM1* exon 9, a knockout line derived from H9 human embryonic stem cells, and retinal organoids grown from it. A patent is a dated public claim, not a reported measurement, so these are recorded as disclosed construct parameters rather than as measured efficacy. [8,28,29,104]

6.3 Endothelin blockade and IGF1: component and combination effects

Kobayashi and colleagues identified early *Edn2* induction in a light-sensitive *Prom1*-deficient retina and reported attenuation of photoreceptor death, gliosis and vessel narrowing with the receptor-selective antagonists BQ-123 (endothelin receptor A) and BQ-788 (endothelin receptor B), given intraperitoneally. The subtypes were not interchangeable across readouts. Photoreceptor death fell with BQ-123, with BQ-788 and with both together. Both antagonists reduced photoreceptor death on that endpoint; their relative effect sizes remain untested. They diverged elsewhere: BQ-788 reduced the number of GFAP-positive glial cells and BQ-123 did not, while BQ-123 promoted recovery of retinal vessels. Receptor subtype is therefore a variable in its own right. This is a direct *PROM1*-model intervention antecedent rather than a general vascular analogy. The Japanese research reports document the provenance of this programme; the controlled experimental evidence assessed here comes from the later experiments. [49,54]

Shigesada and colleagues then used bosentan, a dual endothelin-receptor antagonist, and AAV-mediated Igf1 expression, both delivered intravitreally. The two studies therefore differ in agent, receptor selectivity and route, and their results are related, not replications of one another. The vector was AAV2 driven by a cytomegalovirus promoter, which carries no photoreceptor restriction, and it was delivered intravitreally; the paper's own transduction control found GFP broadly across the retina with strong signals in the ganglion cell layer and in Müller glia. So this experiment leaves open whether IGF1 acts on photoreceptors cell-autonomously, and the paper's own account of death proceeding through cell-to-cell communication points the same way. The components had distinguishable effects. IGF1 delivery increased photoreceptor survival and reduced TUNEL labelling but did not significantly improve ERG responses by itself in the tested regimen. Bosentan reduced gliosis and cell death and improved selected ERG outcomes; the combination produced broader functional improvement relative to the deficient control. The data establish bosentan monotherapy effects and broader

improvement for the combination relative to the deficient control; pharmacological synergy remains a separate question requiring a prespecified interaction contrast. [26]

The AAV cargo in this experiment was Igf1, not *PROM1*. A restored retinal NADH/NAD⁺ ratio supports a changed metabolic state; whether glycolytic flux changed remains open. The mTOR-loss experiment used a Nestin-CreERT2 context affecting both neural and glial populations, so photoreceptor-specific effects remain open. These distinctions matter when assigning the site and mechanism of rescue. [26]

Endothelin is also not uniformly injurious across retinal contexts. Bramall and colleagues ran the experiment both ways in other inherited-degeneration models. Deleting *Edn2* globally increased photoreceptor survival, by 110% in *Pde6b rd1/rd1* mice at postnatal day 15 and 60% in *RHO P347S* mice at day 40, but the same benefit did not appear in retinal explants, and *Edn2*-null mice have systemic abnormalities, so the authors attributed the effect at least partly to systemic loss of function. Restoring *Edn2* locally in photoreceptors by adeno-associated virus then increased survival by 18% at day 14. Opposite manipulations both raising survival are not readily explained by one retinal endothelin effect, and the systemic attribution rests on the explant and systemic findings rather than on the directions of manipulation alone. Dose, compartment and timing could accommodate both directions, so a retinal contribution to the global knockout is not excluded. That paper reports both experiments directly. This does not negate the *Prom1* antagonist results. The direction of intervention may therefore depend on receptor, compartment, dose, stage and local versus systemic exposure. The same caution applies to mTOR, and there the relevant experiment has been run. In a CRISPR *PROM1*-knockout ARPE-19 line, Torin treatment completely inhibited both phosphorylated Akt and phosphorylated S6 ribosomal protein and still had no effect on Atg5 expression. Atg5 abundance is a marker rather than a flux measurement, so what that shows is an unchanged marker under signalling inhibition, not a demonstrated failure to restore autophagic activity. Flux impairment in the retinal pigment epithelium and the requirement for mTOR signalling in other retinal compartments therefore suggest that responses to whole-retina activation or inhibition may differ by compartment. [21,26,53,64]

6.4 Retinoid modulation: direct mouse benefit, a bounded human hypothesis

Dellett and colleagues administered fenretinide from birth in the hybrid *Prom1*-knockout model. They reported structural preservation and improved scotopic ERG, but not significant photopic improvement. This is early preventive/developmental intervention, not demonstrated rescue of established adult cone or macular disease. The study identified reduced *Abca4* and *Rdh12* expression and reported structural and scotopic benefit; increased A2E or lipofuscin remains an unmeasured mechanistic hypothesis. The A1120 comparison in the discussion rests on unpublished observations and therefore supplies a lead for a future head-to-head comparison. [25]

Two human observations refine that hypothesis. The *PROM1*–*ABCA4* family report provides a candidate interaction, whereas Paavo and colleagues did not find generally elevated quantitative short-wavelength autofluorescence in their *PROM1* macular-disease series. Together they favour allele-, background- and stage-specific retinoid investigation and do not establish a universal *ABCA4*-like bisretinoid phenotype. Normal-range quantitative autofluorescence (qAF) does not exclude every retinoid-related process, but it removes a surrogate that would otherwise stand in for an unmeasured one. The source bounds its own result: the atrophic lesions lay within a central zone of under seven degrees eccentricity, while

qAF was measured in segments seven to nine degrees outside the fovea. Its authors state that they therefore cannot exclude initially elevated cone-derived fluorophores in the lesion zone. The normal-range values constrain a currently elevated bisretinoid phenotype in the sampled ring, not the earlier fluorophore burden in the atrophic centre. [56,57]

Tinlarebant's DRAGON programme is an adjacent STGD1 retinol-binding-protein-4 strategy, not a *PROM1* efficacy study. Its clinical development may inform assay selection and mechanism-based trial design, but a Stargardt-like appearance is not sufficient to transfer its disease mechanism. For a *PROM1* retinoid programme, the biochemical subgroup, target engagement and structure–function response remain to be established prospectively. [72]

7. Human studies: genetic relevance without a genotype-specific efficacy estimate

The dated study map separates disease modification, cell-based approaches and optogenetic bypass from observation alone (Table 5). No *PROM1*-specific human efficacy estimate was verified in the records assessed here. This statement concerns the reviewed evidence, not an exhaustive assertion that no unpublished or locally registered programme exists.

EVOLVE (NCT07425574) is observational and was recruiting in the record updated 25 August 2026. The registry permits ABCA4-associated disease or Stargardt-like disease caused by another appropriate macular-dystrophy gene. Astellas explicitly names *PROM1* among the latter examples. The absence of the literal gene name from the registry is consequently not a reason to classify *PROM1* as excluded. Its exclusion of a competing molecular diagnosis addresses confounding or a second disorder, not the entire non-ABCA4 inclusion route. Eligibility remains dependent on phenotype, inheritance and other criteria, and does not establish actual *PROM1* enrolment. [34,66]

STARLIGHT (NCT05417126) explicitly permitted a clinical Stargardt diagnosis or pathogenic ABCA4, ELOVL4 or *PROM1* variants. This six-participant phase 2a optogenetic study was completed in September 2023. Its genetic eligibility is relevant, but a *PROM1* participant or subgroup benefit was not verified. A subsequent planned Stargardt trial has its own recruitment status, separate from the completed study. Separately, Nanoscope's 9 September 2026 announcement reported acceptance and filing of the MOGENRY biologics licence application for severe retinitis pigmentosa (RP); that is a sponsor-reported regulatory milestone, not approval and not *PROM1* replacement. [37,67]

ASP2020 (NCT07734064) adds a newer phase 1b subretinal cell-therapy programme. Its record, updated 16 September 2026, was recruiting and included a non-ABCA4 Stargardt-like genetic route. This creates potential relevance for appropriately characterised *PROM1* macular disease without proving that a *PROM1* participant was enrolled or benefited. It is neither a *PROM1* gene-augmentation trial nor evidence that the initiating compartment has been established. [68]

The broader RP programmes differ in their relevance to *PROM1*. OCU400/liMeliGhT (NCT06388200) was active, not recruiting, in the 31 July 2026 record, with 140 participants and a gene-agnostic arm; its strategy is not *PROM1* replacement. The jCell JC02-88 record was active, not recruiting, and the current sponsor page confirms completed recruitment. Hopkins reports that NAC Attack, the N-acetylcysteine (NAC) trial, completed recruitment in August 2025 with 485 participants. SparingVision reported completed PRODYGY dosing in February

2026, with 27 treated participants and six controls. These programmes may inform preservation strategies in selected RP phenotypes, but they do not automatically cover every *PROM1* macular presentation or supply a *PROM1*-specific effect estimate. [35,36,38,39,71]

For NPI-001, Nacuity’s current programme page identifies NCT07710196 (protocol C-26-01) and says US sites are enrolling. Its inspected registry record, updated 17 July 2026, instead says not yet recruiting. The related NCT07290530 record (C-24-01), updated 1 July 2026, also says not yet recruiting. The records give estimated starts of August and December 2026 respectively. Those dates do not resolve the recruitment discrepancy or establish that one registration superseded the other. The sponsor-linked programme can be identified, but counting the two records as independent trials would need clarification first. Both concern Usher-associated RP rather than general eligibility for isolated *PROM1* disease. [40,69,70]

8. Adjacent preservation approaches: useful experiments at different evidence tiers

Table 4 separates intervention evidence obtained in *PROM1* systems from evidence adjacent to them. The phase 1 oral-NAC RP study provides human functional observations, but its non-randomised design and mixed disease context do not establish *PROM1*-specific preservation. NAD-related work provides separate preclinical leads: retinal metabolism dependent on nicotinamide phosphoribosyltransferase (NAMPT), nicotinamide mononucleotide (NMN) in retinal detachment and nicotinamide riboside in light-induced degeneration concern different insults and models. A 2026 study extends this line, reporting that oral nicotinamide mononucleotide protected rats with retinitis pigmentosa and that NAD⁺ modulated mitochondrial vulnerability in MERTK-associated models, with the effect attributed to a glyceraldehyde-phosphate-dehydrogenase–mitochondria pathway. MERTK governs outer-segment phagocytosis, so that work sits closer to the clearance argument above than the earlier NAD reports do. It remains a different gene and a different insult. Convergence on redox or energy metabolism is a rationale for testing target engagement in *PROM1* tissue, not a substitute for that test. [30–33,75]

Siqueira and colleagues’ small multiwavelength photobiomodulation study in RP used 590, 660 and 850 nm light and reported a short-term acuity change in a single-arm design, so durability against a sham comparator is open. Visual-field outcomes did not provide matching evidence of improvement. Learning, regression to the mean, masking and follow-up duration therefore remain material to interpreting the result. This is not a direct *PROM1* experiment, and wavelength names alone do not specify a transferable retinal dose. [41]

Exercise studies in rd10 and inducible I307N-rhodopsin models make activity a credible experimental variable, but genotype, parental exposure, offspring development and timing of light challenge differ. A ketogenic, low-protein diet in rd10 is likewise a combined nutritional intervention, but one of its components has already been varied between arms. Both ketogenic arms induced systemic ketosis and only the low-protein arm increased photoreceptor thickness, electroretinogram amplitudes and optokinetic thresholds. The arms were not matched on exposure: beta-hydroxybutyrate was higher in the protective arm by 30%, 43% and 56% at postnatal days 30, 40 and 50, so a ketone threshold or an interaction remains open and the study concludes that the role of ketosis is unclear. In wild-type mice the same diet reduced electroretinogram amplitudes at the maximum flash intensity by about 20% at postnatal day 30 and about 30% at day 50, a progressive decline rather than a fixed offset, which is a

measured retinal effect and not only a nutritional confounder. Protein restriction, energy intake, altered growth and other metabolic effects remain confounded with one another. These studies motivate controlled comparisons with nutrition and exposure measured. [42,43,47]

One asymmetry in this section is worth stating. These exposures differ from a drug in that every patient is already receiving a dose. Light, diet, sleep timing and activity are not interventions a person can decline; someone living with *PROM1* disease has all four today, set without guidance. The absence of an evidence-based recommendation therefore does not leave that person at a neutral baseline, only at an unexamined one. This is an argument for measuring how much light contributes to progression. It is the open quantity behind Section 5.1, where dark-rearing protection in one mouse line is set beside an outer-segment defect that precedes light exposure. Someone whose retina is exposed to light every day has to decide whether to act as if ordinary illumination modifies this disease, and how. It is equally an argument against issuing an unevidenced prescription, since a specific wrong exposure is no improvement on an unexamined one. The experiments in Section 10.4, on graded light exposure and the separable biochemistry of retinoids, and in Section 10.7, on clock, activity, diet and photobiomodulation, are designed to replace the default with a measured quantity, which is what would let a patient and a clinician choose on evidence rather than on plausibility.

Sleep and circadian biology require the same separation. An RP sleep-quality association does not establish that sleep loss accelerates *PROM1* disease. Genetic disruption of a retinal clock in a rhodopsin-disease context is not equivalent to experimental sleep deprivation. Deleting *Bmal1* from the retinal pigment epithelium blunted the daily phagocytic peak while leaving the aggregate twenty-four-hour phagosome count, which its authors report as the total number engulfed, without a significant difference between genotypes, and the same study reported no detectable retinal or RPE degeneration by electroretinography, optical coherence tomography, optomotor response, morphology or autofluorescent particle analysis in mice examined at six and eighteen months of age, the older group some sixteen months after the clock was disrupted. That is a dissociation rather than an absence of evidence: the morning peak at zeitgeber time 1 (ZT1), one hour after lights on, was blunted while the nocturnal peak at ZT14 was not affected, the twenty-four-hour phagosome total did not differ significantly, and no detectable degeneration followed. What was tested is a change in the daily profile with the aggregate count unchanged, not the absence of rhythm. A standing phagosome count is a burden rather than a rate, so this separates timing from total burden without establishing that degradation throughput was unaffected. Losing the daily peak is therefore not sufficient on its own to cause degeneration over that interval. It bounds, without excluding, a model in which mistimed clearance becomes harmful only where structural reserve is already reduced, which is where the question now sits. [44–46]

9. Natural history: no single progression rate

9.1 Layer thinning, atrophy expansion and residual cones

Table 6 collects the clinical cohorts and the natural-history endpoints taken from each, with the boundary on every one. Großpötl and colleagues provide the two-year OCT coefficients, and the cohort they report is ProgStar-4, described in Section 9.2: their methods cite its design paper, and their eligibility and five sites are its own. Spectral-domain optical coherence tomography was that study's secondary endpoint, so these are its secondary-endpoint results at 24 months. The body reports 15 enrolled participants, 13 with matched gradable right eyes and seven with matched outer-ring data; the abstract reports 13 enrolled. This review uses each denominator in its own context. Selected annual thickness coefficients were $-1.550 \pm 0.481 \mu\text{m}$ for the outer-ring outer nuclear layer (ONL), $-0.714 \pm 0.238 \mu\text{m}$ for outer-ring inner segments and $-1.939 \pm 0.511 \mu\text{m}$ for inner-ring RPE. The uncertainty attached to these slopes is the reported standard error, not the baseline standard deviation. [48]

The original genotype supplement further limits transfer of these slopes. Of the 13 analysed participants, ten have c.1117C>T listed without a second variant, two have two other variants listed, and one has a different, incompletely specified cDNA entry. A blank second-variant field does not independently document zygosity or phase. This is therefore a variant-mixed, predominantly p.Arg373Cys-listed cohort, not a pure biallelic-null series; its pooled slopes are therefore not recessive or allele-specific progression estimates. [48]

Großpötl and colleagues' cohort also illustrates floor effects. Central inner-segment thickness was already zero in seven of 13 eyes and outer-segment thickness in nine of 13. A structurally meaningful lesion cannot continue to register thinning below zero, so a flat central slope may coexist with ongoing disease elsewhere. Selected coefficients with nominal significance across many layer–region comparisons are useful exploratory estimates, not automatically validated surrogate endpoints or sufficient inputs for a powered efficacy trial. Repeated-measure covariance, segmentation repeatability, baseline reserve and missingness all bear on how such coefficients would perform as trial endpoints. [48]

The intact-area measurements require similar care. Intact area is how much of a mapped region still has a layer measurable above zero, in square millimetres; mean thickness is that layer's depth averaged over the whole region, with lost tissue counted as zero. A patch that thins without shrinking lowers thickness alone, and one that retreats at its edge lowers both. Selected annual intact-area coefficients are $-0.062 \pm 0.017 \text{ mm}^2$ for the outer-ring ONL and $-0.397 \pm 0.132 \text{ mm}^2$ for inner-ring RPE, with $-0.114 \pm 0.005 \text{ mm}^2$ for inner-ring inner segments as printed; the reported standard error and p value for that last coefficient are difficult to reconcile, as Table 6 records. The thickness and intact-area estimates concern different measurements, and a shared OCT origin does not make them combinable. [48]

Ricca and colleagues add a dominant p.Arg373Cys macular-disease series of 15 patients from five kindreds, with different atrophy patterns and a reported overall expansion rate of $0.36 \pm 0.32 \text{ mm}^2/\text{year}$, expressed as mean $\pm$ standard deviation. That rate comes from the 12 eyes of six patients with imaging separated by at least twelve months, not from the full 15-patient cohort, and it pools geographic-atrophy area for some phenotypes with ellipsoid-zone-loss area for others. The phenotype-specific variation makes atrophy geometry and starting stage important. This is not the same endpoint as loss of intact photoreceptor area, nor a rate to apply to all recessive or dominant *PROM1* patients. Family clustering and the longitudinally gradable subset bear on any subsequent reanalysis. [62]

Two cross-sectional studies place these cohort-specific slopes against common measurements. Quantified spectral-domain optical coherence tomography in 3,534 patients across 176 genes showed recessive *PROM1* disease with higher ellipsoid-zone and retinal-pigment-epithelium loss overall, while in dominant disease both were confined to the centre. The significance figures for that dominant-versus-recessive contrast are reported jointly for BEST1 and *PROM1* rather than separately, so what transfers is the direction and distribution of loss rather than a p value; the study is also a preprint whose cohort overlaps the UK service population already cited. Separately, outer retinal tubulation was present in eight of 14 *PROM1* patients in a multicentre series that found it in 104 of 561 patients across thirteen genes, and that series groups *PROM1* with genes whose lesions show chorioretinal atrophy rather than with photoreceptor-specific retinitis pigmentosa. It probably overlaps the Taiwanese series, so the transferable content is the feature and its frequency within the gene. [91,92]

Collison and colleagues' four-patient recessive series describes a recognisable combination that the earlier recessive reports had not emphasised: oval macular lesions, small circular hypoautofluorescent lesions in the posterior pole on near-infrared autofluorescence, extramacular circular atrophy in three of the four, markedly reduced cone-isolated electroretinogram amplitudes with variably reduced rod-isolated amplitudes, and a dark-adapted combined waveform resembling a normal rod-isolated response. The same paper notes that a similar combination occurs in cone-rod dystrophy caused by CDHR1. All four patients were compound heterozygous with segregation confirmed in three families. [93]

Yates and colleagues followed six individuals with recessive *PROM1* disease for a median of 11.8 years, range 0.5 to 25.5, with symptom onset at 7.2 years. Contraction of the subfoveal ellipsoid zone length preceded moderate visual loss, printed there as best-corrected acuity LogMAR < 0.6, and severe loss, by three to five years. That lead time is the longest structural anticipation of acuity loss in this review. Eye-specific contraction slopes are given in the caption to their Figure 4 for the four patients with more than three visits, and they vary by an order of magnitude: 199.7 and 107.1 μm a year in the two eyes of one patient, against 16.7 and 34.5 in another, with R^2 from 0.40 to 0.99. Eight eyes from four patients are not eight independent observations, and the fitted periods differ, so these are individual fitted slopes rather than a pooled progression rate. Central subfield thickness was given, declining in all four at per-eye rates of 1.6 to 6.5 μm a year, with R^2 from 0.42 to 0.98, and the authors describe it as a gradual decline beside the ellipsoid-zone contraction rather than an equivalent measure. Six patients from one Australian programme do not establish an age-specific forecast for every biallelic genotype. Schließleder and colleagues' four-subject deep-phenotyping series adds multimodal and adaptive-optics evidence about residual cone structure, complementing rather than replacing longitudinal functional measures. [58,63]

9.2 A treatment-aligned endpoint hierarchy

A preservation study requires viable target tissue and an outcome capable of changing over its observation interval. An early photoreceptor strategy may use residual ellipsoid-zone extent, cone structure and local sensitivity; an atrophy-edge strategy may favour lesion expansion with spatially matched function. A bypass strategy in advanced disease instead asks whether surviving circuitry can support improved visual performance, and need not imply restoration of normal outer segments. In the same four recessive patients described by Collison and colleagues, Park and colleagues recorded pupillary light responses despite absent or markedly attenuated full-field electroretinograms. The reviewed clinical and experimental studies support this separation of treatment aims [8,9,48,58,62,63,67]. Where the measurement sits also

matters. Kawashima and colleagues looked for the first detectable lesion in a database of 209 Japanese patients with central retinal dystrophies, separately for the retinal pigment epithelium on autofluorescence and for the photoreceptor layer on optical coherence tomography. A lesion small enough to fix a starting point was found in 34 eyes on autofluorescence and in 16 eyes on tomography, and 82.4% and 93.8% of those respectively lay between 500 and 1,000 micrometres from the foveal centre, a ring they name the parafoveal fragile zone. Three of their 89 genetically solved families carried *PROM1*, too few for a genotype-based comparison, so this is context from neighbouring dystrophies rather than a *PROM1* result. It bears on endpoint design all the same. The zone sits inside the inner ETDRS ring, which runs from 0.5 to 1.5 millimetres from the foveal centre, and is barely over a third of that ring by area, so a ring-averaged thickness dilutes an early change in it with the quieter annulus around it. [106,122]

The proposed primary endpoint is selected before treatment assignment or unmasked follow-up, with repeatability measured in the intended genotype–stage stratum. Acuity, microperimetry, low-luminance performance and imaging relate spatially and longitudinally; they are not interchangeable confirmations. Square-root or effective-radius transformations may reduce some dependence on lesion size, but their suitability depends on lesion geometry, multifocality and the biological question. One prospective *PROM1*-specific natural-history study has already been completed on this pattern. ProgStar-4 enrolled 15 patients aged six and over with disease-causing *PROM1* variants and specified ocular lesions, at five centres in the United States and Europe, with visits scheduled every six months over two years. Its primary longitudinal measure was growth of atrophic lesions on fundus autofluorescence, with rate of retinal thinning and photoreceptor loss on spectral-domain optical coherence tomography, retinal sensitivity on microperimetry and best-corrected acuity by the Early Treatment Diabetic Retinopathy Study protocol as secondary measures. Its published abstract gives the baseline distribution: 10 of 29 eyes had definitely decreased autofluorescence, with a mean lesion area of $3.2 \pm 3.5 \text{ mm}^2$ and a range of 0.36 to 10.39 mm^2 , and mean mesopic posterior-pole sensitivity was $8.8 \pm 5.8 \text{ dB}$. The study ran from December 2014 to March 2018. Its 24-month spectral-domain optical coherence tomography results are the coefficients already set out in Section 9.1, because that report is this cohort followed to two years. No progression report for the primary endpoint was available, so atrophy growth on autofluorescence has a defined starting distribution here and no measured rate. [48,96,97]

A wider *PROM1*-specific natural-history effort, building on that endpoint set, would harmonise these choices across the reported families and identify overlap among them.

10. Seven experimental programmes

The following programmes are proposals, not experiments performed for this review. Table 7 gives the same seven as a grid of decisions, prespecified contrasts and the safeguard each contrast needs. Each links an existing observation to a comparison capable of changing the therapeutic conclusion. Their priority is translational leverage: a successful assay identifies which treatment, construct, compartment or patient stratum deserves the next experiment.

10.1 Dominant allele: augmentation, suppression or both?

The discriminating comparison sets an endogenous heterozygous p.Arg373Cys model against a heterozygous null, a biallelic deficient model and corrected controls, with arms for wild-type augmentation across a dose range, allele-selective suppression, and suppress-and-replace. Patient-derived work of this kind depends on independent donors and multiple corrected or reverse-knock-in clones. Wild-type and mutant RNA and protein are separable quantities, and reading them separately alongside membrane localisation, outer-segment structure, survival and function is what makes the comparison interpretable. The transgenic dominant precedent and deficiency-rescue studies define the starting evidence, not the answer. [2,5,8,9]

The key comparison is whether augmentation alone achieves meaningful functional rescue in the endogenous mutant without reducing mutant abundance. A phenotype worse than that of a matched heterozygous null supports a mutant-specific effect, but does not uniquely establish dominant-negative interference. Improvement after suppression is also compatible with toxic gain of function. Mechanistic discrimination requires stoichiometric interaction and rescue experiments, not a single treatment response. A narrow confidence interval excluding a prespecified meaningful benefit is more informative than a non-significant p value from an underpowered study.

10.2 Isoform and glycosylation: total effect versus per-protein activity

This question turns on sequence-verified retinal and comparator *PROM1* constructs under the same promoter, vector backbone and dose series, with untranslated regions stated explicitly. The total effect is estimated at equal vector input, followed by an abundance-matched secondary comparison. Transcript stability, splice products, glycosylation, trafficking and functional rescue are separable by orthogonal detection methods, and a maturation time course with controls for epitope-dependent signal loss keeps the abundance estimate from resting on a single epitope. [7,9,19]

One half of that design has been built before, though not for isoforms. Yang and colleagues' study, the one that identified p.Arg373Cys, built four transgenic lines on one 4.4-kilobase rhodopsin promoter and reported each line's expression as a multiple of endogenous mouse Prom1: the two wild-type lines at one and two fold, and the two mutant lines at eight and two fold. Rate of photoreceptor loss tracked expression level within the mutant lines, and neither wild-type line degenerated. In the matched pair, a wild-type line and a mutant line were both at two fold and only the mutant caused disease, so the injury is not a consequence of transgene dose alone. That contrast bears against a pure dose artefact without settling the question, because no wild-type line was tested at the eight-fold level reached by the more severe mutant line, and the mutant lines themselves show a dose response. [2]

Equal localisation and equal maturation, if imposed on the primary analysis, may remove the mechanisms through which a construct works. A construct that rescues by improving abundance or trafficking remains therapeutically informative even if its abundance-matched activity is similar. Conversely, an expression increase without structural or functional benefit does not by itself make a construct the better therapy. This design also separates the effect of isoform choice from the combined promoter and coding-sequence changes in some disclosure comparisons. [28,29]

10.3 Compartment and clearance: photoreceptor, RPE or both?

Reciprocal, temporally controlled photoreceptor- and RPE-selective perturbations, with independent verification of cell distribution, address this directly. In co-culture, a two-by-two design combines mutant and corrected photoreceptors with mutant and corrected RPE. Outer-segment production, binding, engulfment, degradation and validated autophagic flux are separable steps, and a single readout conflates them. Compartment-selective rescue and spatial mapping are informative before extensive tissue loss and much less so afterwards. Existing RPE flux and regional knockdown studies justify this programme, and the functional clearance evidence they supply is already available. [10,21,24,53]

An RPE rescue that preserves photoreceptors despite persistent photoreceptor *PROM1* deficiency would identify a modifiable propagation component, not necessarily remove the primary structural defect. Conversely, photoreceptor rescue with persistent RPE dysfunction may reveal a later durability limit. Factorial and temporal comparisons can estimate these contributions without forcing the disease into an exclusively photoreceptor-first or RPE-first model.

10.4 Light and retinoids: photic load and biochemical mechanism are separable

A design crossing genotype and genetic background with defined cyclic illumination and graded light exposures, alongside a separately controlled retinoid intervention, separates the photic and biochemical contributions. Treatment after disease onset answers a different question from early prevention. Retinal exposure, temperature, circadian phase and adaptation are measurable quantities, and scotopic and photopic outcomes behave differently. Direct retinoid/bisretinoid assays and quantitative autofluorescence accompany structural rescue; they are not inferred from it. [23,25,57]

The informative result is whether a genotype–background subgroup shows a measured biochemical abnormality that tracks target engagement and functional preservation. Rescue without the proposed biochemical change redirects the mechanism; it does not establish A2E involvement. A normal-range human autofluorescence phenotype also makes biomarker-negative models or patient-derived cells an important comparator.

10.5 Bosentan and IGF1: proposed tests of the interaction and the therapeutic window

A factorial design with vehicle, bosentan, IGF1 and combination arms, matched delivery controls and a prespecified interaction contrast on a clinically interpretable outcome scale would test the interaction. Local exposure and treatment after degeneration begins are separate conditions, and receptor-specific engagement, glial activation, vascular changes and photoreceptor function are separately measurable. The receptor-subtype differences already reported in this model also make a selective-antagonist arm informative alongside the dual antagonist. A separate dose-window experiment examines whether sustained IGF1 exposure preserves benefit or introduces later injury. The existing component effects and adjacent-model endothelin counterevidence motivate this design. [26,54,64]

When exposure is shared, the animal is the randomisation unit; when local treatments differ, an explicitly modelled paired-eye design preserves the denominator. Animals and eyes are different denominators, and reporting them separately keeps that difference visible. Histological survival and ERG rescue are separate endpoints, because the original IGF1 monotherapy result shows that one need not imply the other. A positive combination-versus-control contrast

establishes combination efficacy in that setting; only the prespecified interaction analysis addresses more-than-additive benefit.

10.6 Modifiers: a controlled two-locus and multi-donor test

The informative donors are those sharing a *PROM1* allele but differing in trajectory, including a specifically characterised ABCA4 candidate interaction where one is available. A two-locus isogenic design compares *PROM1* alone, ABCA4 alone, both and neither, while a multi-donor replication estimates background dependence. Phase, transcript effect and the possibility of a second diagnosis all bear on whether an allele is a modifier at all, and each remains open until it is established. Mature photoreceptor–RPE systems can then link retinoid handling, membrane phenotype and survival. [12,25,56,57]

Donor is the unit for inference about human genetic backgrounds; clone, organoid, field and cell are nested observations. A robust genotype interaction reproduced across donors would justify a stratified clinical hypothesis. A single rescued clone is still useful but cannot establish a population effect. Independently ascertained families and prospective measurements would test whether a clinical interaction holds across the severity range, which retrospectively selected severe cases alone cannot establish.

10.7 Clock, activity, diet and photobiomodulation: what has and has not been separated

These candidates are separate hypotheses, and component controls are what allow an effect to be attributed to any one of them. A factorial design estimates component effects and their interaction together, which is what the bosentan and IGF1 programme in Section 10.5 proposes. Two have already been separated in part, and the remaining questions are narrower than the group as a whole suggests. Daily profile and total phagosome burden came apart in the retinal pigment epithelium clock knockout: the morning peak at ZT1 was blunted, the nocturnal peak at ZT14 persisted, and the twenty-four-hour phagosome total did not differ significantly, so the two are separately measurable and a change was demonstrated only in the daily profile. Degradation rate is a third quantity that a phagosome count does not measure, and it bears on whether mistimed clearance is harmful. In diet, ketosis and protein restriction were varied together rather than separated in rd10 mice: both ketogenic diets induced systemic ketosis, only the ketogenic low-protein arm preserved photoreceptors, and that arm also reached higher ketone concentrations, so the design does not isolate either variable. What is still unseparated is more specific: for diet, protein restriction against energy intake and weight trajectory; for activity, parental exposure, growth and photic load. Photobiomodulation requires sham exposure, masked outcome assessment and defined retinal dosimetry. [41–47]

A staged programme can use target engagement to discontinue ineffective conditions before a longer preservation study, while requiring replication of any promising signal in a *PROM1*-relevant model. A promising signal would go to independent replication, with randomisation, a primary outcome and effect-size intervals prespecified. Combination studies are informative when they carry their own component arms and matched exposure controls, so that individual effects and their interaction can be told apart.

11. Discussion and priorities

11.1 Priorities: which comparisons matter most?

The most consequential advance is the transition from a structural disease description to experimentally demonstrated rescue in deficient systems. The largest remaining transfer is to endogenous dominant disease and to human stage-specific targets. Isoform processing, RPE clearance and the photoreceptor–RPE interface give concrete reasons why a vector’s presence alone is an inadequate efficacy criterion. Endothelin, IGF1 and retinoid results identify modifiable injury pathways, but their direction and clinical relevance depend on compartment, exposure and allele. [7–10,21,25,26,53–57]

The criterion used here is which single result would most change what is done next. On that criterion the highest-priority comparisons are endogenous dominant strategy testing, which is the dose series against a heterozygous null in Section 10.1; construct-controlled rescue, which is the equal-input then abundance-matched comparison in Section 10.2; and compartment-resolved intervention delivered after degeneration has begun, which is the reciprocal photoreceptor and retinal pigment epithelium test in Section 10.3, with treatment stage carried as a variable there and in the light and pharmacological programmes that follow it. Natural-history work is integral to these priorities, because it determines whether the rescued process can be measured in the intended patients. The accompanying tables organise variants, models, interventions, human programmes, cohorts, experimental decisions and publication-linked research expertise around those questions.

11.2 Recurrent alleles and evolutionary explanations

Table 8 maps the contributions demonstrated in the cited publications, as a route to the groups already working on each question. The recurrence of p.Arg373Cys and its occurrence on differing haplotypes invite population-genetic questions. The haplotype observation is specific: the positional-cloning study that identified the allele compared disease haplotypes across the STGD4 family, the MCDR2 family and a third dominant cone-rod family, found them distinct, and concluded that the mutation arose independently in each. The Japanese series reached no such conclusion, reporting instead that its genetic data were too limited for haplotype analysis to distinguish single from multiple founder events. Those are different evidential positions, and neither alone discriminates recurrent mutation from older shared ancestry, recombination or incomplete haplotype resolution. A CpG-compatible substitution is a mechanistic mutation hypothesis, not a measured mutation rate. A founder or selection claim requires appropriate population denominators, ancestry-aware haplotypes, demographic modelling and independent fitness-related evidence. Clinical ascertainment cannot supply those quantities. [2,4,6,13,15]

Likewise, *PROM1*’s functions outside the retina, including reported interaction with VEGF, establish biological pleiotropy rather than a selective advantage for a retinal-disease allele. Discordant developmental or intestinal phenotypes across *Prom1* model reports and the subsequent commentary also make allele construction and background relevant to their interpretation. These observations are useful constraints on evolutionary explanations; they do not by themselves establish that a particular disease variant has been positively selected. [50–52]

11.3 Evidence access and remaining uncertainty

This review is limited by unequal access to final methods and supplements, small or overlapping cohorts, and incomplete genotype-specific reporting in broad clinical programmes. Some clinical sources are included qualitatively, without a complete patient-level roster in this review. Appendix A summarises source access and verification. The findings converge in direction; a numerical synthesis here would imply independence, precision or coverage that the available records do not establish.

Shoukat and colleagues' September 2026 genotype–phenotype preprint is a retrospective database study of 190 *PROM1* variants drawn from ClinVar and the Human Gene Mutation Database, analysed by contingency tables and a Monte Carlo chi-square test of independence, alongside a separate case series of seven unrelated patients from one ocular-genetics practice. Its database arm therefore shares sources with the cohorts reported elsewhere in this review rather than adding independent observations, and it is not merged with them here. Its reported associations run the same way as the allele classes set out in Section 3.1: dominant missense variants with milder macular disease, truncating frameshift and nonsense variants with recessive inheritance and more widespread degeneration. [74]

12. Conclusion

PROM1-associated retinal degeneration is best approached as an allele-, compartment- and stage-dependent set of disorders. Replacement in deficient systems is the strongest direct therapeutic foundation; dominant-mutant rescue, construct processing and the durability of photoreceptor–RPE repair remain decisive translational questions. Direct RPE-flux, endothelin-intervention and retinoid-modulation evidence broadens the experimental options, while human autofluorescence and natural-history findings constrain overgeneralisation. The studies proposed here aim to connect a controlled mechanism to sustained function in an allele- and stage-matched model and to a measurable outcome in the corresponding clinical subgroup.

Data and materials

This paper is self-contained: the nine evidence tables, the reference list and the source-access summary in Appendix A are part of it. It contains no new individual-patient data and no raw-omics reanalysis, and third-party full texts are not redistributed with it.

The deposit also carries an accessible HTML edition, a spoken-text version written to be read aloud, a description of the figure for readers who cannot see it, a narrated audiobook, and a data file listing the 40 smaller multi-gene cohorts summarised in Section 3.3, each with its size, its *PROM1* figure in the unit it reports, and any overlap with other cohorts. The HTML edition holds the same text, tables and references as this PDF. The spoken text and the audiobook carry the narrative through to the end of the conclusion, without the tables or the reference list.

The paper and its figure are licensed under [CC BY-NC 4.0](#), which permits copying, sharing and adaptation with attribution for any non-commercial purpose. Non-commercial academic, clinical and patient-education reuse is welcome under that licence and needs no further permission. Anything the licence does not cover, including publication under CC BY where a journal requires it, can be asked for; permission will normally be given.

Appendix A. Source access and verification

This appendix says what was in front of the author for each source, so that a reader can weigh a claim against it. The review cites 162 sources, and the full published text was read for 133.

Fifteen are not journal articles or preprints but trial registry entries, sponsor announcements and study protocols: 34 to 40, 66 to 72 and 97. Wherever one is used the paper gives the kind of record and the date it carries.

Eight are known only from an abstract or a supplement: 16, 22, 60, 95, 103, 113, 117 and 118. What is claimed from them is held to what that abstract or supplement reports, which is a statement about what was available here rather than about the quality of the study. The article for 117 is not open, and its supplementary table supplied the *PROM1* rows used here; for 95 the supplements supplied the variant list. Reference 9 was read as a preprint, and its journal version was seen only as an abstract, as its entry records.

Six could not be obtained: 126, 134, 140, 147, 149 and 157. All six are multi-gene cohorts in Table 9, and each row there says what could not be confirmed. Nothing is claimed from them beyond what an abstract or a registry entry states.

Every paragraph that carries a citation, and every row of the nine tables, has been checked sentence by sentence against the sources it cites. Where a claim rests on material that could not be obtained, the table row or the sentence says so.

A further version is planned, and closing these gaps is the first thing it would do. Anyone willing to send a copy of a paper listed above would be helping; the address is under Correspondence and corrections.

AI assistance

Artificial-intelligence systems were used throughout this work and their role is stated here in full.

OpenAI GPT-6, Anthropic Claude models (Opus 5, Fable 5.1) and xAI Grok 4.6 were used for literature retrieval, for organising and cross-checking extracted evidence, for independent adversarial review of successive drafts, and for drafting and revising prose. Their findings were consolidated into a disposition log retained by the author.

Claims were not accepted on a single model's assertion and were reviewed in multiple passes by a variety of agents tasked with a variety of approaches and focuses. Human review was the ultimate arbiter. Fifteen sources were audited claim by claim against their full text, and Appendix A records for every reference whether it rests on full text, an abstract or a registry record, and which of those consulted have been through that audit.

The review reports and the disposition log are preserved by the author, who monitored and guided the work, read the whole paper, set its final architecture, directed the revisions, and takes responsibility for the integrity and accuracy of the work as a whole, including any remaining errors.

Competing interests

The author declares no financial competing interests: no funding, employment, consultancy, patent, shareholding or other financial relationship with any person or organisation with an interest in this work.

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Each version carries its own number, printed on the title page and in the footer of every page. This deposit holds the current version, so a citation that carries the version number identifies which text is meant, which matters here because corrections are invited and Appendix A names the gaps a later version would close.

Correspondence and corrections

If you have published, preprinted or unpublished work on *PROM1*, prominin-1 or CD133 in the retina, including negative results, corrections to anything stated here, or a cohort or dataset not represented, the author would be glad to receive it at cjb.research@sciworlds.com and to cite it in a later version.

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Evidence tables

PROM1-associated retinal degeneration · 21 September 2026 · Version 1.90

Reference numbers correspond to the manuscript. These are selected evidence maps, not pooled datasets. A publication, family, individual, eye, donor, clone and cell are different units. Where a cell says a thing is not confirmed or not verified, that describes this review's source assessment, not proof of absence. Registry status is tied to the displayed record date; sponsor statements are separately attributed.

Bhattacharya and colleagues' review synthesises photoreceptor and RPE biology and the proposed connection with atrophic AMD. Independent-experiment counts in these tables derive from the primary studies. [11]

Table 1. Illustrative allele classes and what they establish

Allele or class	Human/model context	Established observation	Therapeutic inference and boundary	References
p.Arg373Cys	Dominant macular and broader retinal phenotypes; classic transgenic mouse precedent	Clinical segregation and experimental disruption of outer-segment morphogenesis	Distinguishing augmentation, suppression and combined treatment for this allele context requires an endogenous heterozygote; a transgene is not an endogenous knock-in.	[2,4,5,15]
Recessive truncating/splice alleles	Often earlier, more widespread retinal disease	Clinical deficiency-oriented phenotype; molecular effects differ by transcript and allele	Replacement is strongly motivated, but biallelic inheritance does not prove every product is a complete null.	[1,5,6,18]
p.Leu618Arg	Recessive missense example, reported homozygous	Included in the recessive clinical literature	Residual function remains an open question; missense alleles are not uniformly dominant.	[5]
p.Arg373Cys plus c.1142-1G>A	Italian father–son/maternal segregation	Son with the additional maternal splice allele was more severely affected	Human allelic-interaction lead, not proof of dominant-negative action; previously reported family.	[61]
c.1354dupT, source-reported	Patient iPSCs; three patient backgrounds, two corrected lines	Molecular repair and genetically informative cellular comparators	Mature retinal functional rescue and multi-donor replication remain separate tests.	[12]
c.1645-1648del plus c.2321delC	Compound heterozygous in a six-year-old proband; constructs transfected into HEK293T cells	Reduced protein stability and altered subcellular localisation; a possible impairment of cell viability, which its authors put as the variant “might impair” it; no change in apoptosis	The journal abstract reports an elevated LC3-II to LC3-I ratio, read there as autophagic activation, and the conference abstract of the same work reports the ratio in the opposite direction; one static ratio does not fix flux direction either way.	[22]
Deep-intronic/pseudoexon allele, c.2077-521A>G	Consanguineous family, two affected siblings; RAB28 and <i>PROM1</i> both exon-negative; minigene splice assay	Pseudoexon integrated into the mutant transcript, giving a premature termination codon and a presumed functional null	Transcript-aware interpretation and allele-specific splice repair are plausible. Assignment is by minigene rather than patient transcript, and clinical use depends on an exact transcript and genome build.	[65,93]
Intronic deletion, intron 21	Recessive cone–rod disease	Published noncoding disease mechanism	Distinct from the pseudoexon allele; repair designs are not interchangeable. Assigned by minigene after RT-PCR of <i>PROM1</i> in blood and fibroblasts was negative for the region tested.	[73]

Table 1 (continued)

Allele or class	Human/model context	Established observation	Therapeutic inference and boundary	References
<i>PROM1</i> with an ABCA4 candidate modifier	Small human pedigree	Reported phenotype modification in a two-gene context	Candidate interaction, not a quantified population modifier effect or universal ABCA4-like mechanism.	[56,57]
Exon 15 deletion, chr4:15992516–15997089 (GRCh38)	Three families in one whole-genome cohort	Heterozygous with a Stargardt-like phenotype in one family; homozygous or compound heterozygous with retinitis pigmentosa in two others	A structural variant is a class in its own right, and the same deletion did not behave the same way in every family.	[89]
c.221-20T>G, intron 2	Two homozygous Italian siblings, diagnosed Stargardt disease	Complete exon 3 skipping shown by RT-PCR on whole-blood complementary DNA, p.Asp74AlafsTer1	Assigned from patient RNA rather than minigene; the accessible isoform carries the affected exon, which is not true of every <i>PROM1</i> region.	[88]
‘c.1697dupA; p.Asn566Lysfs*2’	Homozygous in one examined Saudi Arabian woman; affected brother by history	Childhood cone–rod disease, high myopia, nystagmus; online 2014, before the Polish c.1697dupA report	Same allele in separate settings; high myopia was observed in one examined patient	[87,114]
‘c.2327A>T’, p.Asp776Val; ‘c.2485G>A’, p.Asp829Asn	Brazilian macular cases 1 and 2; homozygous and heterozygous respectively	Macular atrophy and computational pathogenicity predictions	No segregation or direct variant-function evidence in the three-case report	[111]
‘c.C1902G(p.Y634X)’ plus ‘c.C1682+3A>G’	Two affected Chinese brothers, compound heterozygous	Nonsense construct was truncated, unstable and mislocalised; intronic minigene impaired splicing	Transfected-cell and minigene effects; patient retinal protein and RNA were not measured	[115]
‘NM_001145851: c.1445dupT (p.F482fs)’	Homozygous in two unrelated Chinese retinitis-pigmentosa families	Recessive segregation; absent in 200 matched controls	No allele-specific functional rescue in this report	[112]
‘c.1909C>T’ (p.Gln637); ‘c.2050C>T’ (p.Arg684)	Two newly reported truncating variants within a six-patient United States series	Truncating consequences reported	Phase, per-variant phenotype and functional effects are not specified	[113]

Reported allele strings have not been globally re-normalised. No transcript accession or genome build is assigned to protein-only entries.

Table 2. Experimental models, assays and transfer limits

Model / source	Perturbation and compartment	Principal use	Replication or interpretation boundary	References
Classic dominant mouse	Mutant <i>PROM1</i> transgene	Disk morphogenesis and mutant interference	Expression/insertion context differs from endogenous heterozygosity.	[2]
<i>Prom1</i> -null mouse backgrounds	C57BL/6 versus C57BL/6×CBA/NSlc	Background-dependent and light-modified degeneration	Dellett's non-histological experiments used the hybrid background, so results from the two backgrounds are not poolable.	[3,23,25]
CRISPR <i>Prom1</i> knockout, C57BL/6	Whole-animal deletion of 184 bases in exon 2	Outer-segment protein abundance and trafficking measured separately, by immunoblot and immunofluorescence	Four animals per group for the imaging and electroretinogram comparisons, not for every assay; abundance and localisation were measured at different ages, so the two time courses are not directly comparable.	[105]
Human retinal splice models	Retinal-exon perturbation and maturation	Isoform-dependent retinal development	Exon skipping, exon-containing RNA and ciliation frequency are distinct outputs.	[7]
<i>PROM1</i> -knockout human retinal organoids	Guide RNA targeting exon 9 in H9 human embryonic stem cells	Disclosed human model used for construct testing	A patent disclosure, used here for the model description; no phenotype data from it are assessed here.	[104]
Patient-derived replacement models	Homozygous p.Glu207Ter organoids; <i>Prom1</i> -null mice	AAV7m8-CRXp-hPROM1 in organoids; AAV8-CRXp-hPROM1 in mice	Final abstract verifies constructs; detailed final methods remain incompletely accessed.	[9]
Human and mouse RPE cultures	Human <i>PROM1</i> reduction; CRISPR loss in immortalised mouse RPE	Flux, lysosomal processing and outer-segment handling	Distinct models; mouse outer-segment feeding did not separate binding, ingestion and digestion.	[21,53]
Regional mouse RPE perturbation	Subretinal AAV2/1 CRISPR delivery	Ability of RPE injury to affect photoreceptors	Scrambled controls and spatial mapping support localisation; delivery bias is not a genetic cell-specific driver.	[10]
<i>Prom1</i> -null <i>Xenopus</i>	Whole-animal deficiency	Deposits, cone-rod disease and RPE atrophy	Whole-animal mutation does not assign cell autonomy.	[24]
<i>prom1</i> -null and <i>cdhr1</i> -null <i>Xenopus</i>	Single and double whole-animal nulls	Outer-segment disk architecture, cone-selective severity and structure against function	Evagination, fusion and disk spacing all persist. Random indels in the F0 single and double nulls; defined deletions in the <i>prom1</i> -null F1 and the <i>cdhr1</i> -null F2 to F3; the F1 electroretinogram differed only in the scotopic B-wave.	[86]
Acute-light single-cell discovery	Two male knockout littermates; one light, one dark	Cell-type candidate localisation	One animal per condition. Many cells do not supply between-animal replication; later experiments are separate.	[26]

Table 2 (continued)

Model / source	Perturbation and compartment	Principal use	Replication or interpretation boundary	References
Cultured mouse RPE transcriptomics	CRISPR knockout clones; batch adjustment and exclusions	Stress/clearance pathway hypotheses	Partial EMT interpretation; no raw reanalysis in this review. GSE298665 is a locator, not a new result.	[27]
<i>PROM1</i> /TTYH1 cultured-cell vesicles	Expression and lipid interaction	Membrane bending and cholesterol association	Biophysical evidence, not demonstration of retinal functional rescue.	[55]
<i>prom1b</i> -null zebrafish	Whole-animal deficiency in the photoreceptor-relevant orthologue	Cone-first degeneration with rods viable to nine months; Prph2 mislocalisation and disrupted oligomerisation	A candidate mechanism for the cone asymmetry, reported in a fish retina.	[94]
Normal and transgenic <i>Xenopus</i> photoreceptors	Antibody and tagged-protein localisation in rods and cones	Open lamellar rims carry prominin-1 opposite peripherin-2 at closed rims	Localisation, not a disease allele or direct membrane-fusion assay	[119]
Comparative vertebrate retinal expression	Axolotl, zebrafish, chicken and mouse retinal mapping	Inner nuclear layer prominin-1-positive populations outside photoreceptors in zebrafish, chicken and mouse, but not in axolotl	Expression does not establish cell-autonomous disease or a rescue target	[120]

Table 3. *Prom1* mouse studies: background, construction, protein and light

Study and model	Strain background	How the mutation was built	Prominin-1 still made?	Light received	References
Yang et al. 2008: dominant transgenic	C57BL/6	Human p.Arg373Cys <i>PROM1</i> on a rhodopsin promoter, injected into embryos; two lines expressing it at 2 and 8 times the mouse's own level	Yes: the mouse's own prominin-1 is intact and mutant human protein is added	12-hour light/dark cycle; intensity not stated	[2]
Zacchigna et al. 2009: knockout	Congenic C57BL/6, backcrossed from a 129/Sv and Swiss line	Exon 2, with the start codon, replaced by a neomycin cassette in embryonic stem cells	None detected in kidney by immunoblot	Not stated; cell death still rose in mice kept dark to postnatal day 20 (data not shown)	[3]
Dellett et al. 2015: knockout on two backgrounds	C57BL/6, and a hybrid of C57BL/6 and CBA/NSIc; experiments other than histology used the hybrid	RIKEN knockout line CDB0623K, made in an earlier study; hybrid bred by in vitro fertilisation with CBA/NSIc	Not stated in the source	12-hour cycle at 160 lux; darkness below 0.5 lux protected the photoreceptors	[25]
Kobayashi et al. 2021: knockout	Hybrid of C57BL/6 and CBA/NSIc	RIKEN knockout line CDB0623K, also used in [25]; its targeting vector carries <i>lacZ</i>	Not stated in the source	12-hour cycle, below 0.5 lux in the dark phase; dark rearing to postnatal day 21 prevented the rise in <i>Edn2</i> , <i>Bcl3</i> and <i>Gfap</i> expression and gave fewer GFAP-positive glial cells; light tests used 3,800 lux for 3 hours	[54]
Xiao et al. 2021: CRISPR knockout	C57BL/6J, after three generations of backcrossing	CRISPR/Cas9, a 184-base deletion in exon 2	None detected in retina by immunoblot or immunofluorescence	Not stated in the source	[105]
Bhattacharya et al. 2023: cultured mouse RPE	Immortalised RPE from C57BL/6J mice	CRISPR/Cas9 clones, made with a guide RNA targeting exon 5	None detected by immunoblot	Not applicable: cells in culture	[53]
Bhattacharya et al. 2024: regional RPE knockdown	C57BL/6J	Cas9 and a <i>Prom1</i> guide RNA delivered by subretinal AAV2/1 at 4 to 6 weeks of age	Not stated in the source	12-hour light/dark cycle; intensity not stated	[10]
Shigesada et al. 2024: knockout, light challenge	Not stated in the source	RIKEN line with the <i>Prom1</i> locus replaced by <i>lacZ</i> , made in an earlier study	Not stated in the source	Usually a 12-hour cycle; for light tests, dark-reared pups had their eyelids opened and received 15,000 lux for 3 hours	[26]

Table 3 (continued)

Study and model	Strain background	How the mutation was built	Prominin-1 still made?	Light received	References
Yanardag et al. 2024: rd19, with a <i>Prom1</i> -CreERT2 comparison	rd19: mixed C57BL/6J and 129SVE/J carrying the <i>Rpe65</i> leucine-450 allele, and C57BL/6J with methionine-450; CreERT2 line: B6N;129S	rd19, a spontaneous nonsense mutation in exon 9 (p.Lys268Ter); comparison line with CreERT2 in place of <i>Prom1</i>	rd19: none detected in retina by immunoblot or staining; CreERT2 line reported as losing prominin-1 expression	12-hour cycle, or complete darkness from birth for 30 days; darkness did not preserve photoreceptor function in any of the three, nor photoreceptor numbers in rd19	[23]
Huo et al. 2025: cultured mouse RPE	Immortalised RPE from C57BL/6J mice	CRISPR/Cas9 clones KO22 and KO26.4, made as in [53]	None detected by immunoblot	Not applicable: cells in culture	[27]
Ji et al. 2025: knockout, gene replacement	C57BL/6J	CRISPR/Cas9 deletion of about 59.6 kb, from exon 2 to exon 20	None detected by immunofluorescence or immunoblot	12-hour light/dark cycle; intensity not stated	[8]
Holmes et al. 2026, conference abstract: rd19, gene replacement	Not stated in the abstract	rd19, the allele used in [23]	Not stated in the abstract	Not stated in the abstract	[103]
Xu et al. 2026, preprint methods: knockout, gene replacement	C57BL/6JCya	CRISPR/Cas9 deletion of exons 2 to 20	None detected in retina by immunoblot	Not stated in the preprint	[9]

One row for each *Prom1* mouse study in Tables 2 and 4, cultured mouse cells included, by year of publication. Section 5.1 explains how each of the four conditions can change a result. “Not stated” means the source does not give it.

Table 4. Intervention evidence: direct versus adjacent

Intervention	Evidence context	Observed result	What remains unestablished	References
AAV human <i>PROM1</i> , wild-type coding sequence under a short rhodopsin-kinase promoter	<i>Prom1</i> -knockout mice; 1 μ L at post-natal day 14, or 2 μ L at day 21 or 28 after thinning had begun; fellow eye given buffer	Outer nuclear layer preserved on the treated side at the five-month endpoint in both later cohorts, 32.3 to 34.9 μ m against 16.6 to 20.6 in the day-21 cohort and 29.2 to 36.8 against 14.6 to 19.8 in the day-28 cohort; expression widespread across nasal and temporal retina in the day-21 cohort	The later cohorts also received twice the dose, so age is not the only variable. In the day-14 cohort no significant difference in dark-adapted electroretinogram amplitudes was detected at three months and a difference was detected at five, and amplitudes still deteriorated. How much viable tissue is required is not established	[8]
AAV human <i>PROM1</i> under a rhodopsin-kinase promoter	<i>Prom1</i> -knockout (Rd19) mice, injected at post-natal day 15 and at day 30	Scotopic and photopic rescue reported after the day-30 injection, and long-term recovery of scotopic function from one to ten months after the day-15 injection, with outer-segment structure and photoreceptor numbers preserved to ten months	A conference abstract: the injection volume is 1 μ L against saline, but the vector concentration's exponent is not recoverable from the record consulted, and animal numbers and statistics are not given, so the result is treated qualitatively	[103]
Processed/photoreceptor-associated <i>PROM1</i>	Patient-derived and mouse deficiency models	Replacement rescue reported	Unconfounded isoform comparison and final/preprint methods reconciliation	[9]
DNA correction and splice-mechanism evidence	Patient cells for coding alleles; minigene assay or patient RNA for the noncoding alleles, according to whether an accessible isoform carries the affected exon	Molecular correction/splice-mechanism evidence	Sustained mature retinal function and clinical delivery	[7,12,65,73,88]
BQ-123 / BQ-788 (receptor-selective, intraperitoneal)	Direct <i>Prom1</i> -deficient mouse experiments	Attenuated photoreceptor death, gliosis and vessel narrowing; subtype effects differ by readout	Receptor subtype, exposure and translation to a dual antagonist	[54]
Bosentan (dual antagonist, intravitreal)	Direct <i>Prom1</i> -deficient mouse experiments	Reduced gliosis/cell death and selected ERG improvement	Human efficacy, optimal receptor/compartments/exposure and delayed-treatment durability	[26]

Table 4 (continued)

Intervention	Evidence context	Observed result	What remains unestablished	References
AAV-Igf1	Direct <i>Prom1</i> -deficient mouse experiment	Survival and metabolic-state effects; no significant ERG benefit alone in tested regimen	Translation from structural survival to function; quantitative dose window	[26]
Bosentan + AAV-Igf1	Same model	Broader ERG improvement versus deficient control	Formal interaction/synergy; animal-versus-eye accounting	[26]
Fenretinide	Hybrid <i>Prom1</i> -null mice; treatment from birth	Structural/scotopic benefit, not significant photopic benefit	Adult rescue; directly measured bisretinoid mechanism; human subgroup response, noting that macular autofluorescence was not generally elevated in a human <i>PROM1</i> cohort, although it was sampled seven to nine degrees outside the fovea while the lesions lay within seven, leaving the central burden unmeasured [57]	[25]
A1120	Unpublished observation mentioned in Dellelt discussion	Comparative statement, not a separate published efficacy dataset	Quantified reproducible head-to-head effect	[25]
NAC; NAD-directed candidates	Human mixed RP phase 1; distinct preclinical metabolic/insult models	Human functional observations or model-specific protection	<i>PROM1</i> -specific controlled preservation; these are not one interchangeable intervention class	[30–33,75]
Photobiomodulation	Small uncontrolled human RP study; 590/660/850 nm	Short-term acuity change; fields did not supply matching significant improvement	Sham-controlled durable benefit and <i>PROM1</i> -specific response	[41]
Activity / ketogenic low-protein diet	rd10 and I307N-rhodopsin models	Model-specific preservation signals; both ketogenic diets induced ketosis but only the low-protein arm preserved photoreceptors, and that arm reached higher ketone concentrations	Ketone exposure was not matched, so ketosis is not excluded; still to isolate activity, parental exposure, protein restriction against energy intake, and growth effects. The same diet reduced wild-type electroretinogram amplitudes at maximum flash intensity by about 20% at postnatal day 30 and 30% at day 50	[42,43,47]
Sleep / retinal clock	Human RP association and genetic clock models	Sleep-quality association; clock and phagocytosis-timing effects	Sleep-loss causality in <i>PROM1</i> ; whether mistimed clearance becomes harmful in a structurally vulnerable retina	[44–46]

DMSO is a vehicle control in the direct pharmacology, not an active *PROM1* treatment. Bramall reports local protective EDN2 effects in other genotypes, which do not directly test the *Prom1* antagonist findings; the endothelin-protective data come from Bramall, and Dellelt and Shigesada do not report any. [25,26,64]

Table 5. Dated human-study map

Programme / identifier	Type and status evidence	PROM1 relevance	Enrolment / efficacy boundary	References
EVOLVE; NCT07425574	Observational; recruiting, registry update 25 Aug 2026; estimated 90	Registry permits appropriate non-ABCA4 macular genes; Astellas explicitly names <i>PROM1</i>	Potential eligibility, not confirmed <i>PROM1</i> enrolment; no study treatment	[34,66]
STARLIGHT; NCT05417126	Phase 2a optogenetics; completed 28 Sep 2023; record 9 Nov 2023; six actual participants	Registry explicitly names <i>PROM1</i> alongside ABCA4/ELOVL4, with clinical-diagnosis route	Actual <i>PROM1</i> participants and genotype-specific outcomes not verified; future trials are separate	[67]
ASP2020; NCT07734064	Phase 1b cell therapy; recruiting, record 16 Sep 2026; estimated 30	Broader Stargardt-like genetic route can be relevant to <i>PROM1</i> phenotype/inheritance	No confirmed <i>PROM1</i> enrolment or efficacy; not <i>PROM1</i> replacement	[68]
OCU400/liMeliGHT; NCT06388200	Phase 3; active, not recruiting, record 31 Jul 2026; 140 actual	Gene-agnostic RP arm; phenotype and residual tissue matter	Not a generic macular-dystrophy route or <i>PROM1</i> augmentation; subgroup benefit not verified	[36]
NAC Attack; NCT05537220	Phase 3; coordinating centre reports recruitment completed Aug 2025, 485 participants	Broad RP preservation rationale	No verified <i>PROM1</i> -specific effect estimate; not an open recruitment listing	[35]
PRODYGY/SPVN06; NCT05748873	Phase 1/2; sponsor 9 Feb 2026 reports dosing complete: 27 treated, six controls	Cone-preservation mechanism; entry depends on protocol stage/criteria	Sponsor milestone, not a <i>PROM1</i> result or current open-enrolment assertion	[38]
jCell/JC02-88; NCT06912633	Active, not recruiting, record 30 Jul 2026; current sponsor confirms recruitment complete	Broad RP genetic scope, with functional eligibility constraints	The current sponsor page describes 6.0 million cells	[39,71]
SLO-RP Preserve; NCT07710196; C-26-01	Phase 3; registry 17 Jul 2026: not yet recruiting; current sponsor: US sites enrolling	Usher-associated RP, not isolated <i>PROM1</i> eligibility	Sponsor explicitly identifies this record; estimated Aug 2026 start does not establish recruitment.	[40,69]
NPI-001; NCT07290530; C-24-01	Related phase 3 record; 1 Jul 2026: not yet recruiting; estimated Dec 2026 start	Usher-associated disease	Relationship to C-26-01 unresolved; the records do not establish supersession, and do not establish two independent trials either.	[70]

Table 5 (continued)

Programme / identifier	Type and status evidence	PROM1 relevance	Enrolment / efficacy boundary	References
MOGENRY/MCO-010; RESTORE programme	Sponsor 9 Sep 2026: BLA accepted/filed for severe RP	Optogenetic bypass of lost photoreceptors	A pre-approval optogenetic bypass, with efficacy in the <i>PROM1</i> subgroup still unverified	[37]
Tinlarebant/DRAGON; NCT05244304	Adjacent STGD1 RBP4-targeting programme; historical HRA protocol summary	Mechanistic comparator, not a verified <i>PROM1</i> trial	The historical protocol summary does not establish current status, and a Stargardt-like presentation does not establish <i>PROM1</i> efficacy	[72]
ProgStar-4; NCT02410122	Completed prospective observational study, December 2014 to March 2018; actual enrolment 15	<i>PROM1</i> -specific by eligibility; fundus-autofluorescence lesion growth as the primary longitudinal measure	Observational, so no efficacy question arises. Its 24-month secondary-endpoint OCT results are reference 48; no progression report for the primary endpoint was available	[48,96,97]

Current sponsor pages were accessed 18 September 2026. Where a row's status evidence is a sponsor or coordinating-centre statement, the registration number identifies the programme and the registry record itself was not separately inspected for that row. Registry "last update posted" dates are not the same as review-access dates. A site can have screening restrictions beyond a broad genetic inclusion route. The map is not an exhaustive national-registry census.

Table 6. Clinical cohorts and selected natural-history endpoints

Evidence source	Denominator / setting	Contribution	Boundary for inference	References
United Kingdom and German centres	<i>PROM1</i> series across two specialist clinics; separate IRD service cohort	Inheritance and molecular ascertainment	Service ascertainment is not population prevalence.	[5,14]
Japan; Korea	Japan: 10 patients/8 families; Korea: 7/5	Phenotype breadth and R373C variability	Mixed classifications; each cohort discloses one patient reported earlier (Japan, same group 2016; Korea, Kim and colleagues' 2017 p.Arg373Cys report), so adding the two series double-counts those individuals if the two earlier reports are counted as well.	[6,13,116]
China; Taiwan; India	<i>PROM1</i> , Stargardt or RP ascertainment; the Indian 49-patient Stargardt cohort included one patient with a <i>PROM1</i> variant	Geographic and phenotypic breadth	Different sampling frames; recessive carriers do not establish dominant penetrance.	[15–18]
Saudi case, Khan and Bolz	One examined patient/one family; brother affected by history	Homozygous c.1697dupA with high myopia and childhood nystagmus	Published online before the Polish observation; not a cohort frequency	[87,114]
Brazilian macular series, Salles	Three men selected from 25 records	Two proposed <i>PROM1</i> alleles and one molecularly unresolved case	Three variant carriers are not three established <i>PROM1</i> diagnoses; patient-level overlap with Porto [109] is unreported	[109,111]
Chinese recessive cases, Liu	Two patients/two unrelated families	Homozygous c.1445dupT and severe young-adult retinitis pigmentosa	Separate from the cited Chinese dominant series; no pooled Chinese frequency	[112]
Chicago pupillometry, Park	Four adults/three families; same four as Collison	Pupillary responses persisted when full-field electroretinograms were absent or markedly reduced	The same four patients as Collison [93], so the two reports are not independent observations	[93,122]
Japanese supplement, Suga	Five patients/four pedigrees in the supplement	Four of five patient rows attribute prior reporting to Fujinami [6]	Four of the five were reported earlier; the fifth patient's prior reporting is unresolved	[6,117]
Shanghai conference report, Chen	33 patients, 16 dominant/17 recessive; family count unstated	Earlier, more widespread recessive disease and optical-coherence-tomography differences	Patient and variant roster and overlap are not reported	[118]
Single-institution series, Lee JJ	Six patients, ages 12–47; family count unstated	Recessive versus dominant imaging phenotype; two truncating variants	Variant allocation and phase per patient, and overlap, are unreported	[113]
Spain	Dominant/recessive clinical series	Expanded phenotypes	Exact splice-variant roster not listed here	[60]
Italy	10 patients; previously reported father–son pair	Mixed dominant/splice genotype	The previously reported father–son pair is one family in this series.	[61]

Table 6 (continued)

Evidence source	Denominator / setting	Contribution	Boundary for inference	References
Großpötlz: sample	15 enrolled; 13 matched right eyes; 7 outer-ring eyes	Two-year OCT trajectories	Abstract says 13 enrolled; body supplies detailed accounting.	[48]
Großpötlz: variants	Original supplement: 13 analysed patients	10 R373C-only listings; 2 other pairs; 1 incompletely specified entry	Phase and zygoty are not stated for every participant; not a recessive-only slope estimate.	[48]
Outer-ring ONL	7 matched eyes	$-1.550 \pm 0.481 \mu\text{m}/\text{year}$; $p = 0.018$	Slope $\pm$ SE; selected nominal result, not validated surrogate	[48]
Outer-ring IS	7 matched eyes	$-0.714 \pm 0.238 \mu\text{m}/\text{year}$; $p = 0.024$	$n = 7$ for this endpoint; the cohort n of 13 does not apply.	[48]
Inner-ring RPE	13 matched eyes	$-1.939 \pm 0.511 \mu\text{m}/\text{year}$; $p = 0.003$	Different layer/region from other thickness measures	[48]
Inner-ring IS area	13 matched eyes	Printed $-0.114 \pm 0.005 \text{ mm}^2/\text{year}$; $p = 0.043$	SE and p value as printed; their correspondence is unresolved	[48]
Central OCT floor	13 eyes	IS zero: 7/13; OS zero: 9/13	Flat zero-bound slopes do not exclude progression elsewhere.	[48]
Ricca: atrophy	15 patients/5 kindreds; R373C; rate from 12 eyes of 6	$0.36 \pm 0.32 \text{ mm}^2/\text{year}$ (mean $\pm$ SD)	A mixed endpoint: geographic-atrophy area for some phenotypes and ellipsoid-zone-loss area for others. Geometry and longitudinal subset matter; not a thickness slope.	[62]
Yates: recessive	6 individuals; median 11.8 years, range 0.5 to 25.5; 4 with more than three visits	Subfoveal ellipsoid-zone contraction precedes moderate and severe acuity loss by three to five years; central subfield thickness declines in all four, at per-eye rates of 1.6 to 6.5 μm a year	Eye-specific slopes in the Figure 4 caption range from 16.7 to 199.7 μm a year across eight eyes of four patients, so no pooled rate follows. One Australian programme, six patients; not a universal age-based prognosis	[58]
Schließleder	4 subjects; multimodal imaging	Residual-cone assessment	Structure alone does not establish functional rescue.	[63]
Polish national cohort	190 families screened on a 108-gene panel, 115 solved; three <i>PROM1</i> families	All three ended at retinitis pigmentosa with macular involvement from three different initial labels; one homozygous c.1697dupA case	A panel of genes known in 2013 with a 0.5% frequency filter, raised to 3.0% for <i>ABCA4</i> ; a referral cohort, not a population denominator	[87]
Biallelic versus monoallelic series	Four biallelic and six monoallelic individuals, one centre	Mean onset 9 against 37.3 years; mean acuity 20/200 against 20/30; photopic electroretinogram extinguished or severely attenuated in 4/4 biallelic, normal in 4/5 monoallelic tested	Small and single-centre; the monoallelic diagnoses are listed as seven entries against six patients	[90]

Table 6 (continued)

Evidence source	Denominator / setting	Contribution	Boundary for inference	References
ProgStar-4 base-line	15 patients, 29 eyes, five centres in the USA and Europe	10 of 29 eyes with definitely decreased autofluorescence, mean lesion area $3.2 \pm 3.5 \text{ mm}^2$, range 0.36 to 10.39; mean mesopic posterior-pole sensitivity $8.8 \pm 5.8 \text{ dB}$, reported for 27 of the 29 eyes	Baseline distribution for the cohort whose 24-month OCT results appear in the Großpötl row above; these are not two independent studies	[48,96,97]
Paavo: qAF	Cohort 18 patients/36 eyes; qAF analysed in 12 patients/20 eyes, 18 of 20 within the healthy 95% CI; other imaging results reported over 38 eyes	qAF not generally elevated	The report uses 36 eyes for the cohort and 38 for other imaging analyses; each denominator is carried separately here. The abstract and results both give 18 of 20; the Figure 3 caption gives two of 17 outside the interval; the relationship between these denominators is not stated	[57]
Portuguese dual-inheritance registry	40 genes searched, 22 present, nine showing both patterns across 102 families and 141 patients	Within <i>PROM1</i> , 76.9% of cases recessive, spanning retinitis pigmentosa, macular dystrophy and cone-rod dystrophy	A referral-centre proportion from one country, not a population estimate; read at abstract level	[95]
Cross-sectional quantified OCT	3,534 patients across 176 genes	Recessive <i>PROM1</i> disease with higher ellipsoid-zone and retinal-pigment-epithelium loss overall; dominant disease confined to the centre	Significance reported jointly for BEST1 and <i>PROM1</i> , so the direction transfers and a p value does not; the cohort overlaps the UK service population	[92]
Outer retinal tubulation	14 <i>PROM1</i> patients within 561 across thirteen genes	Present in 8 of 14; <i>PROM1</i> grouped with genes whose lesions show chorioretinal atrophy	Probably overlaps the Taiwanese series, so the feature and its within-gene frequency transfer, not an independent denominator	[91]

ONL, outer nuclear layer; IS, inner segments; OS, outer segments; RPE, retinal pigment epithelium; EZ, ellipsoid zone; qAF, quantitative autofluorescence; SE, standard error; SD, standard deviation. The displayed OCT rows are selected, not every nominally significant result in the paper. Positive atrophy expansion and negative intact-area change are not interchangeable rates.

Table 7. Experimental decisions and prespecified contrasts

Programme	Decisive comparison	Primary interpretive safeguard	Decision enabled	Starting evidence
1. Dominant strategy	Endogenous p.Arg373Cys versus heterozygous null; augmentation, suppression, both	Mutant-specific injury is not uniquely dominant-negative; wild-type and mutant abundance measured separately	Whether augmentation alone merits development	[2,5,8,9]
2. Isoform/processing	Same promoter/backbone; equal-input dose series, then abundance-matched analysis	Localisation, abundance and maturation can be mediators	Total functional rescue discriminates cargo; epitope intensity does not	[7,9,19,28,29]
3. Compartment/clearance	Reciprocal PR/RPE perturbation and two-by-two co-culture rescue	Ingestion, digestion, flux and tissue initiation are separable steps	Which compartment(s) require durable correction	[10,21,24,53]
4. Light/retinoids	Genotype × background × exposure; measured retinoids; early and delayed treatment	Rescue does not establish A2E involvement, and lux does not transfer without retinal dosimetry	Which biochemical subgroup responds, and in what window	[23,25,57]
5. Bosentan/IGF1	Factorial component/combo design and interaction contrast	Eyes nested in animals; structure and ERG distinct; local/systemic context	Component efficacy, interaction and exposure window	[26,54,64]
6. Modifiers	<i>PROM1</i> × candidate ABCA4 in isogenic and multi-donor settings	Donor-level inference; phase and second-diagnosis checks	Whether a modifier justifies clinical stratification	[12,25,56,57]
7. Adjacent approaches	Clock, activity, dietary-component and sham-PBM effects are separable	Integrated daily clearance versus phase; growth and exposure controls	Which component merits direct <i>PROM1</i> preservation testing	[41–47]

All seven are proposed work rather than completed studies, and none carries an effect-size threshold here. A threshold depends on the endpoint, its floor effect and the independent biological unit, which Section 9 and Table 6 set out for the endpoints that have measured rates. A non-significant superiority test is not an equivalence test. Randomisation, masking, attrition and multiplicity handling bear on interpretation before treatment effects are inspected.

Table 8. Publication-linked research expertise

Author cluster	Demonstrated contribution in cited work	Useful research connection	References
Michaelides, Cehajic-Kapetanovic, Birtel, Charbel Issa, MacLaren and collaborators	Dominant/recessive clinical phenotyping and molecular interpretation	Allele-aware cohorts and endpoint design	[4,5]
Fujinami, Tsunoda, Miyake, Iwata and the Japan Eye Genetics Consortium	Japanese national phenotype spectrum, foveal sparing and an accompanying population literature review	Consortium-scale ascertainment for an allele-specific study, with existing links to the United Kingdom cohort	[6]
Großpötl, Schießleder and collaborators	Longitudinal OCT and deep retinal imaging	Segmentation, floor effects and residual-cone measures	[48,63]
Ricca, Han, Stone, Sohn and collaborators	Dominant <i>PROM1</i> macular-atrophy phenotypes	Atrophy geometry and family-aware natural history	[62]
Bhattacharya, Yin, Huo, Chaum and collaborators	Human/mouse RPE autophagy, regional injury and transcriptomics	Clearance assays and reciprocal compartment rescue	[10,21,27,53]
Dellett, Sasai, Shigesada, Kobayashi and collaborators	Background/light-sensitive models; endothelin and IGF1 experiments	Controlled pharmacology, light exposure and component comparisons	[25,26,54]
Moya-Molina and collaborators	Human retinal splicing and maturation models	Isoform-controlled organoid work	[7]
Puertas-Neyra, Coco-Martin and collaborators	Patient iPSCs and c.1354dupT repair	Multi-background and corrected cellular comparisons	[12]
Ji, Ni, Wang, Zhou, Sun; Xu and collaborators	Recent <i>PROM1</i> -delivery studies: Ji (journal article) and Xu (preprint, with the published journal abstract)	Construct comparison, paired fellow-eye design and post-onset treatment	[8,9]
Holmes, Yanardag, Kolandaivelu, Ramamurthy and collaborators	<i>Prom1</i> -knockout mouse biology and a further gene-augmentation programme	A further laboratory report of post-onset benefit, at conference-abstract level, and outer-segment structure as an outcome	[23,103]
Bell, Chao and collaborators	<i>PROM1</i> /TTYH1 membrane and extracellular-vesicle biophysics	Lipid interaction and protein-level mechanism	[55]
Carr, Moritz and collaborators	<i>Xenopus</i> outer-segment architecture, cone-rod and RPE pathology	Non-mouse structural/compartment comparison	[24,86]
Paavo, Lee, Sparrow and collaborators	Quantitative autofluorescence and candidate ABCA4 interaction	Human biochemical stratification	[56,57]
Lu, Tang, Liu and collaborators	Zebrafish <i>prom1b</i> and the Prph2 relationship	Cone-versus-rod mechanism and a second non-mammalian model	[94]

Table 8 (continued)

Author cluster	Demonstrated contribution in cited work	Useful research connection	References
Collison, Fishman, Allikmets and collaborators	Recessive <i>PROM1</i> imaging phenotype and second-allele discovery by whole-genome sequencing	Recognisable phenotype and structural-variant ascertainment	[93]
Strauss, Scholl and the ProgStar-4 Study Group	Prospective <i>PROM1</i> -specific natural history and endpoint selection	Multicentre protocol design and baseline measures	[96,97]
Cremers, Roosing, Banfi and collaborators	Whole-genome sequencing for noncoding and structural alleles in unsolved disease	Patient-RNA transcript evidence and reanalysis of filtered variants	[88]

This is a non-exhaustive map of contributions demonstrated by the cited publications, not a ranking, current-employment directory, endorsement or assertion of resource availability. Affiliations and materials access are recorded in the corresponding publication and the current institutional records.

Table 9. *PROM1* in multi-gene inherited retinal disease cohorts

Cohort and setting	Tested, in the study's own unit	<i>PROM1</i> reported	Boundary for inference	References
Branham et al. 2025, International	33,834 patients from 27,561 families, all submitted with a reported genetic cause	377 patients assigned <i>PROM1</i>	FFB consortium and sponsored testing overlap possible with Mansfield 2020 US registry; individual linkage not supplied. Not confirmed by the available source material: the number tested	[123]
Lin et al. 2024, United Kingdom	3,953 families; 4,415 individuals with molecular diagnoses	reported without a stated count	Same Moorfields service as Pontikos 2020 [14], with later accrual; cannot add allele count to [14] family count	[124]
Perea-Romero et al. 2021, Spain	3,951 index cases/families with DNA, from 4,403 unrelated families and 6,089 affected people; <i>PROM1</i> -tested subset not stated; 2,100/3,951 genetically characterised solved	24 genetically characterised families: 14 non-RP (5 AD, 9 AR) and 10 RP (1 AD, 9 AR)	institutional overlap with Del Pozo-Valero series likely, identities unlinked	[125]
Australian Inherited Retinal Disease Registry and DNA Bank 2022 annual report, Australia	3,429 affected participants with DNA; exact <i>PROM1</i> -sequenced denominator not stated; registry holds 3,894 affected of 9,350 total participants; 1,091 solved	4 families with <i>PROM1</i> assigned causative gene; 4 affected participants with <i>PROM1</i> assigned causative gene	Cumulative registry snapshot; possible patient overlap with Australian cohort papers and Hitti-Malin international maculopathy studies; earlier AIRDR reports are version snapshots, not independent cohorts	[126]
Karali et al. 2022, Italy	2,790 patients; 2,036 patients with potentially conclusive molecular diagnosis	20 cases	Italian centre; patient linkage to Italian <i>PROM1</i> -specific series not stated	[127]
Motta et al. 2018, Brazil	1,246 patients seen; 559 tested; 400/559 solved	9 affected patients with a positive <i>PROM1</i> test result	Brazilian cohort; overlap with Porto 2017 [109] not established at patient or centre level	[128]
Suga et al. 2022, Japan	1,210 pedigrees; 448 pedigrees (37%) solved	4 pedigrees; 5 patients	Four of five <i>PROM1</i> patients (TM-18, TM-127, TM-128, TM-164) cite Fujinami et al. 2020 [6]; TI-32 does not. Not confirmed by the available source material: the number tested, the number solved, the overlap with other cohorts	[117]
Panneman et al. 2023, International RP/LCA/unspecified retinal dystrophy referrals	1,192 probands analysed; 566 very likely solved plus 100 possibly solved	2 probands in the unspecified retinal dystrophy diagnostic subgroup; total <i>PROM1</i> cohort count not printed in main text	International cohort; individual links to national <i>PROM1</i> series not provided	[129]

Table 9 (continued)

Cohort and setting	Tested, in the study's own unit	<i>PROM1</i> reported	Boundary for inference	References
Matczyńska et al. 2024, Poland	1,005 patients, 889 IRD and 116 optic neuropathy; 646 definitive molecular diagnoses	9 patients, counted from distinct identifiers in the supplementary material	Eight patients had prior unsolved exome testing in the same group's earlier project; possible overlap with Matczyńska 2024 Genes 77-patient cohort, but its reported <i>PROM1</i> causal count is zero; separate from Tracewska 2021 laboratory	[130]
Weisschuh et al. 2024, Germany	1,000 probands; 574 definite genetic diagnoses	3 diagnosed probands in gene-by-phenotype table	German service; individual relationship to German <i>PROM1</i> -specific publications unresolved	[131]
Jespersgaard et al. 2019, Denmark	677 individuals; 323 plausible molecular diagnoses	5 individuals, IDs 175, 177, 178, 179 and 440	Three further individuals, IDs 114, 270 and 281, carry a <i>PROM1</i> variant among their additional variants rather than as the assigned gene	[132]
Schlottmann et al. 2023, Argentina	637 families; 773 patients; 379 families solved	5 family IDs (124, 133, 223, 309, 332), one positive patient row each; 5 positive patients represented by those IDs	National Argentine sample; no overlap with cited <i>PROM1</i> -specific series documented	[133]
Conti et al. 2024, Western Switzerland	629 individuals from 490 families underwent molecular analysis; full clinically affected registry 899 individuals from 690 families; 400 molecularly diagnosed individuals from 285 families	Family count not separately stated in accessible main text; Figure 3 family bar not resolved; 17 molecularly diagnosed individuals (14 AD and 3 AR), main Results; 4.3% of 400 solved individuals	Single-centre Western Swiss registry may contain earlier cases in Schorderet 2013 Sion or other Swiss reports; linkage not given	[134]
Heutinck et al. 2024, Netherlands	624 registered patients; 473 genetically examined patients; 76% of 473 examined patients had a genetic cause identified	1 newly resolved on 2022 reanalysis; total <i>PROM1</i> cohort count and variants unconfirmed	Dutch registry may contain patients from earlier Dutch <i>PROM1</i> work; linkage not stated	[135]
Testa et al. 2025, Italy	580 genetically diagnosed patients, all with non-syndromic retinitis pigmentosa	7 <i>PROM1</i> patients; zero had cystoid macular oedema	Italian cohort could share patients with Karali 2022 and other single-centre series; centre and identities not yet reconciled	[136]
Ellingford et al. 2016, United Kingdom-led world-wide clinical referrals	537 patients; 271/537 confirmed or provisional molecular diagnoses	8 individuals	Worldwide referral service and 105-gene panel may overlap other clinical service cohorts; identities unavailable	[137]
Lieviant et al. 2025, Singapore	506 probands; 49.2% of 506 probands solved; Supplementary Table 1 includes solved and probably solved categories	4 unrelated probands by study kinship pruning; two c.1238T>A/c.139del cases explicitly in trans in two families	Singapore cohort; no link to a cited <i>PROM1</i> series stated	[138]

Table 9 (continued)

Cohort and setting	Tested, in the study's own unit	<i>PROM1</i> reported	Boundary for inference	References
Coco-Martin et al. 2021, Spain	488 IRD patients; 129 genetically tested (58 RP, 71 others); 92/129 solved	1 genetically diagnosed patient in autosomal dominant subgroup	Regional Spanish cohort; individual linkage to [60] or Madrid cohorts not stated	[139]
Sanders and Votruba 2025, United Kingdom	403 patients; 166 patients with definitive causative variant	2 individuals	Welsh clinic distinct from UCL/Moorfields; individual overlap not established	[140]
Hitti-Malin et al. 2022, Netherlands-led macular-disease cohort	372 probands with BAM files; 42 of 110 very likely or possibly solved	3 probands	prior reporting not stated in the source; 070872 c.1767G>A is VUS	[141]
Ganapathi et al. 2022, United States	357 unrelated individuals; 84/357 confirmed plus 22/357 likely solved	At least 2 unrelated probands; family count not explicitly assigned to <i>PROM1</i> ; 2 confirmed <i>PROM1</i> molecular-diagnosis probands in Supplementary Information 3; one additional likely diagnosis with a homozygous <i>PROM1</i> VUS and additional secondary/possible variants	US tertiary cohort; no patient identity linkage to other US <i>PROM1</i> series seen	[142]
Arai et al. 2015, Japan	349 probands	3 unrelated families with c.1738A>C reported; three c.1738A>C families documented but Table 2 states four <i>PROM1</i> probands and conflicts	Japanese <i>PROM1</i> review [6] discusses p.Asn580His and regards it as likely benign; 2015 RP cases also have EYS/CRB1 variants, so causal attribution is uncertain	[143]
Priglinger et al. 2024, Germany	342 completed reports among 443 referrals (101 pending); analytic age-stratified cohort 309 patients; 270/342 definitive diagnoses; 309 patients in analytic cohort, including possibly solved cases	4 patients: <i>PROM1</i> (1) aged 0-6 and <i>PROM1</i> (3) aged 7-17	Single-centre Munich pediatric cohort; no overlap established with the other cohorts here	[144]
Mishra et al. 2024, United Kingdom	325 selected patients	30 patients with autosomal dominant <i>PROM1</i> -associated macular dystrophy, 16 male and 14 female	Same Moorfields service as Pontikos 2020 [14] and Lin 2024; substantial cohort overlap plausible, individual linkage not supplied. Not confirmed by the available source material: the number tested	[145]

Table 9 (continued)

Cohort and setting	Tested, in the study's own unit	<i>PROM1</i> reported	Boundary for inference	References
De Geer et al. 2025, Sweden	324 unrelated index individuals; 177/324 index individuals (55% rounded) solved	2 unrelated index families; 2 index individuals	May overlap Swedish retesting cohort of Areblom Stanton 2023; shared region/individuals not documented	[146]
Jeon et al. 2024, South Korea	318 NGS-tested patients; 133 diagnosed RP	1 RP patient	Different centre from Hwang et al. 2023 [13]; patient identity not checked	[147]
Chen et al. 2021, Taiwan	312 families/probands; 178/312 families with disease-causing genotypes solved	5 families; 5 probands in Supplementary Data 1; full affected-person count unknown	Later Taiwanese <i>PROM1</i> -specific clinical series may draw from this population; individual linkage not checked	[148]
Patel et al. 2016, Saudi Arabia	292 families	2 families, IDs 09DG00585 and 12DG0402; At least 3 affected individuals: father and son in 09DG00585 footnote, plus sporadic case 12DG0402; no <i>PROM1</i> patient total printed	Prior reporting of these families is unresolved; identical c.2130+2 deletion notation in Porto [109] does not establish patient overlap between the Saudi and Brazilian cohorts	[149]
Areblom Stanton et al. 2023, Sweden	279 patients; 182 patients (65%) solved	6 patients (RP67LU, RP198LU, RP29LU, RP32LU, RP31LU, RP47LU)	Shared variants and registry leave possible related patients; family count and overlap with other cohorts not stated	[150]
Liu X et al. 2024, China	271 patients; 646 persons including relatives; 34 of 271 patients solved	4 families (FM112, FM157, FM289, FM297); 3 affected patients in FM112 plus at least one proband in each of FM157, FM289, FM297; total affected <i>PROM1</i> patients not stated	Chinese cohort could share recruitment with Chinese <i>PROM1</i> series [15]; individual identity not established	[89]
Surl et al. 2024, South Korea	264 unrelated patients; 166 initial plus 22 newly diagnosed on reanalysis, 188 total	1 unrelated proband/family; 1 patient, GH_0131	Korean cohort; no patient linkage to Korean <i>PROM1</i> series [13] stated	[151]
Birtel et al. 2018, Germany	251 patients; 185 patients (74%) solved	6 patients (Supplementary Table 1 IDs 106–110, 185)	Bonn team and several variants also appear in <i>PROM1</i> series [5], which cites this paper; identical patients are possible but not established by identifiers	[152]
Harford et al. 2026, Ireland	232 patients with macular or cone dystrophy; detailed non-ABCA4/BEST1 subset 72; 35 of 72 initially unresolved subset solved on further testing	3 patients	Irish registry and UK collaborator pool may overlap published <i>PROM1</i> clinical series; patient identity not stated	[108]

Table 9 (continued)

Cohort and setting	Tested, in the study's own unit	<i>PROM1</i> reported	Boundary for inference	References
Tracewska et al. 2021, Poland	190 families after exclusions; 115 families solved	3 families (IDs 394, 400, 422); 3 probands shown in Appendix 3; affected relatives not enumerated for <i>PROM1</i>	Possible overlap with earlier Polish cone-rod series for c.1612delT; identity unresolved	[87]
Zenteno et al. 2020, Mexico	143 unrelated probands; 95/143 probands solved	2 unrelated probands/families	prior reporting not stated in the source; unrelated probands in this cohort	[153]
Britten-Jones et al. 2025, Australia	140 participants; 124 unrelated families; 84 participants from 71 families with probable molecular diagnosis	One case, distinct-family total not separately printed; 1 participant, case 36	VENTURE registry may contain longitudinal <i>PROM1</i> cases; no patient-level overlap established	[154]
Eisenberger et al. 2013, Germany and other referrals	126 patients; 88/126 patients with causative mutation detection	1 patient/family, ID 55	prior reporting not stated in the source	[155]
Ng et al. 2022, China	125 unrelated subjects: 123 sporadic patients (4 STGD, 119 RP) plus 2 Stargardt-family probands	3 sporadic subjects carrying <i>PROM1</i> variants as printed: patient STGD-1, c.1196G>A (p.R399H) and c.1520T>A (p.F507Y), both heterozygous; RP-017, c.857A>G (p.Q286R); RP-048, c.868A>C (p.S290R)	Three variant carriers, not three confirmed <i>PROM1</i> -solved diagnoses. Phase for patient STGD-1 and zygosity for the two RP subjects are not stated in these tables, and pathogenicity is unresolved. The source's 28/125 subjects carrying potential disease-causing variants span six genes and are not a solved count. No stated link to the <i>PROM1</i> -specific Chinese series [15]	[156]
Khan et al. 2017, United Kingdom	115 patients; 45 certain molecular diagnoses; 13 inconclusive; 57 without a causative result	1 <i>PROM1</i> -assigned patient in 45 certain diagnoses, inferred from figure and stated less-than-4% frequency among remaining genes; plus 1 inconclusive patient ID 1796	Same Moorfields service as Pontikos et al. 2020 [14]; likely nested, patient identity not established	[157]

Table 9 (continued)

Cohort and setting	Tested, in the study's own unit	<i>PROM1</i> reported	Boundary for inference	References
Heutinck et al. 2025, Iran	111 patients; 66 patients solved	1 patient	prior reporting not stated in the source	[158]
Jaju et al. 2026, South India	109 unrelated probands; 113 affected participants including 4 relatives; 42 solved plus 8 possibly solved probands	2 diagnosed probands	Article explicitly contrasts with South Indian panel study by Sen et al. [17]; no shared patient linkage stated	[159]
Sen et al. 2023, India	107 patients; 72 patients with pathogenic mutations	5 <i>PROM1</i> -labelled patients with causal certainty varying; RP-7 VUS and RP-53 secondary VUS	Sankara Nethralaya group also appears in Indian Stargardt study [16]; patient overlap not reported	[17]
Jiman et al. 2020, United Kingdom	106 patients; 52/106 probable genetic diagnoses	1 patient with homozygous <i>PROM1</i> variant	UK service; no individual link to Moorfields [14] or other <i>PROM1</i> series stated	[160]
Ge et al. 2015, United States	105 probands; 52/105 probands solved	1 proband/family, N6+A.15	National network cohort; prior reporting not stated in the source	[161]
Diñeiro et al. 2020, Spain	100 patients; 45/100 patients genetically diagnosed	1 index case, OFTALMO.010; 1 patient	Spanish cohort; no explicit linkage to Spanish <i>PROM1</i> series [60]	[162]
Pontikos et al. 2020, United Kingdom	not reported; 3,195 families; 4,236 affected individuals solved	37 families; 53 affected individuals	Same Moorfields service as the later Moorfields cohorts in this table; UK contributors to [6] may also overlap; individual identity not established	[14]

Counts are given in the unit each study prints, and are not converted or pooled: patients, probands, families, pedigrees, index cases and alleles are different units. Several cohorts draw on the same services or reuse patients from earlier reports, as the boundary column records, so no total is given.